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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Coordination Compounds

JEE Main Chemistry · Previous Year Questions with Answer Key

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Numerical	82 questions
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Numerical 82 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Number of paramagnetic complexes among the following is ____ .

$[MnBr_4]^{2-}$, $[NiCl_4]^{2-}$, $[Ni(CN)_4]^{2-}$, $[Ni(CO)_4]$, $[CoF_6]^{3-}$, $[Fe(CN)_6]^{4-}$, $[Mn(CN)_6]^{3-}$, $[Ti(CN)_6]^{3-}$, $[Cu(H_2O)_6]^{2+}$, $[Co(C_2O_4)_3]^{3-}$

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

An excess of $AgNO_3$ is added to 100 mL of a 0.05 M solution of tetraaquadichloridochromium (III) chloride. The number of moles of AgCl precipitated will be ____ $\times 10^{-3}$.

(Nearest integer)

Open solution

Q3. JEE Main 2026 (Online) 5th April Evening Shift

The total number of unpaired electrons present in the d^3, d^4 (low spin) d^5 (high spin), d^6 (high spin) and d^7 (low spin) octahedral complex systems is ____ .

Open solution

Q4. JEE Main 2026 (Online) 2nd April Morning Shift

5.33 g of $CrCl_3 \cdot 6H_2O$, which is a 1 : 3 electrolyte, is dissolved in water and is passed through a cation exchanger. The chloride ions in the eluted solution, on treatment with $AgNO_3$ results in 8.61 g of AgCl. The ratio of moles of complex reacted and moles of AgCl formed is _____ × 10^{-2}.
(Nearest integer)

[Molar mass in g mol^{-1} Cr : 52, Ag : 108, Cl : 35.5, H : 1, O : 16]

Open solution

Q5. JEE Main 2026 (Online) 28th January Evening Shift

The number of isoelectronic species among Sc^{3+}, Cr^{2+}, Mn^{3+}, Co^{3+} and Fe^{3+} is 'n'. If 'n' moles of AgCl is formed during the reaction of complex with formula $CoCl_3(en)_2NH_3$ with excess of $AgNO_3$ solution, then the number of electrons present in the t_{2g} orbital of the complex is _____.

Open solution

Q6. JEE Main 2026 (Online) 28th January Morning Shift

X is the number of geometrical isomers exhibited by $[Pt(NH_3)(H_2O)BrCl]$.

Y is the number of optically inactive isomer(s) exhibited by $[CrCl_2(ox)_2]^{3-}$

Z is the number of geometrical isomers exhibited by $[Co(NH_3)_3(NO_2)_3]$.

The value of $X + Y + Z$ is ____ .

Open solution

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Q7. JEE Main 2026 (Online) 24th January Evening Shift

A chromium complex with a formula $CrCl_3 \cdot 6H_2O$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to $1:2$ electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the process was reacted with excess of $AgNO_3$. The amount of AgCl formed in the above process is ____ g. (Nearest integer)

[Given: Molar mass in $gmol^{-1} Cr: 52; Cl: 35.5, Ag: 108, O: 16, H: 1$]

Open solution

Q8. JEE Main 2026 (Online) 23rd January Evening Shift

Total number of unpaired electrons present in the central metal atoms/ions of

$[Ni(CO)_4], [NiCl_4]^{2-}, [PtCl_2(NH_3)_2], [Ni(CN)_4]^{2-}$ and $[Pt(CN)_4]^{2-}$ is ____.

Open solution

Q9. JEE Main 2026 (Online) 23rd January Morning Shift

The crystal field splitting energy of $[Co(\text{ oxalate })_3]^{3-}$ complex is ' n' times that of the $[Cr(\text{ oxalate })_3]^{3-}$ complex. Here ' n ' is ____ . (Assume $\Delta_0 \gg P$)

Open solution

Q10. JEE Main 2026 (Online) 21st January Evening Shift

Identify the metal ions among $Co^{2+}, Ni^{2+}, Fe^{2+}, V^{3+}$ and Ti^{2+} having a spin-only magnetic moment value more than 3.0 BM. The sum of unpaired electrons present in the high spin octahedral complexes formed by those metal ions is _____ .

Open solution

Q11. JEE Main 2025 (Online) 7th April Evening Shift

The number of paramagnetic metal complex species among $[Co(NH_3)_6]^{3+}, [Co(C_2O_4)_3]^{3-}$, $[MnCl_6]^{3-}, [Mn(CN)_6]^{3-}, [CoF_6]^{3-}, [Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$ with same number of unpaired electrons is _____.

Open solution

Q12. JEE Main 2025 (Online) 7th April Morning Shift

The number of paramagnetic complexes among $[FeF_6]^{3-}, [Fe(CN)_6]^{3-}, [Mn(CN)_6]^{3-}$, $[Co(C_2O_4)_3]^{3-}, [MnCl_6]^{3-}$, and $[CoF_6]^{3-}$, which involve d^2sp^3 hybridization is _____.

Open solution

Q13. JEE Main 2025 (Online) 4th April Evening Shift

A metal complex with a formula $MCl_4 \cdot 3NH_3$ is involved in $sp^3\, d^2$ hybridisation. It upon reaction with excess of $AgNO_3$ solution gives ' x ' moles of AgCl . Consider ' x ' is equal to the number of lone pairs of electron present in central atom of BrF_5. Then the number of geometrical isomers exhibited by the complex is _____.

Open solution

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Q14. JEE Main 2025 (Online) 3rd April Morning Shift

The number of optical isomers exhibited by the iron complex (A) obtained from the following reaction is_____.

$$FeCl_3 + KOH + H_2C_2O_4 \rightarrow A$$

Open solution

Q15. JEE Main 2025 (Online) 2nd April Evening Shift

The spin-only magnetic moment value of M^{n+} ion formed among Ni, Zn, Mn and Cu that has the least enthalpy of atomisation is_____ . (in nearest integer) Here n is equal to the number of diamagnetic complexes among $K_2[NiCl_4], [Zn(H_2O)_6]Cl_2, K_3[Mn(CN)_6]$ and $[Cu(PPh_3)_3I]$

Open solution

Q16. JEE Main 2025 (Online) 2nd April Morning Shift

A transition metal (M) among Mn, Cr, Co and Fe has the highest standard electrode potential (M^{3+}/M^{2+}). It forms a metal complex of the type $[M(CN)_6]^{4-}$. The number of electrons present in the e_g orbital of the complex is _____.

Open solution

Q17. JEE Main 2025 (Online) 29th January Evening Shift

Consider the following low-spin complexes

$K_3[Co(NO_2)_6], K_4[Fe(CN)_6], K_3[Fe(CN)_6], Cu_2[Fe(CN)_6]$ and $Zn_2[Fe(CN)_6]$

The sum of the spin-only magnetic moment values of complexes having yellow colour is _____ B.M. (answer in nearest integer)

Open solution

Q18. JEE Main 2025 (Online) 28th January Evening Shift

The spin only magnetic moment (μ) value (B.M.) of the compound with strongest oxidising power among Mn_2O_3, TiO and VO is _____ B.M. (Nearest integer).

Open solution

Q19. JEE Main 2025 (Online) 28th January Evening Shift

Total number of molecules/species from following which will be paramagnetic is _____.

$O_2, O_2^+, O_2^-, NO, NO_2, CO, K_2[NiCl_4], [Co(NH_3)_6]Cl_3, K_2[Ni(CN)_4]$

Open solution

Q20. JEE Main 2025 (Online) 22nd January Evening Shift

The complex of Ni^{2+} ion and dimethyl glyoxime contains _____ number of Hydrogen (H) atoms.

Open solution

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Q21. JEE Main 2024 (Online) 9th April Evening Shift

Consider the following test for a group-IV cation.

$M^{2+} + H_2S \rightarrow A$ (Black precipitate)+ byproduct

$A +$ aqua regia $\rightarrow B + NOCl + S + H_2O$

$B + KNO_2 + CH_3COOH \rightarrow C +$ byproduct

The spin-only magnetic moment value of the metal complex C is _____ BM (Nearest integer)

Open solution

Q22. JEE Main 2024 (Online) 9th April Morning Shift

Number of ambidentate ligands among the following is _____.

$NO_2^-, SCN^-, C_2O_4^{2-}, NH_3, CN^-, SO_4^{2-}, H_2O.$

Open solution

Q23. JEE Main 2024 (Online) 8th April Evening Shift

Total number of unpaired electrons in the complex ions $[Co(NH_3)_6]^{3+}$ and $[NiCl_4]^{2-}$ is _____.

Open solution

Q24. JEE Main 2024 (Online) 8th April Morning Shift

The 'spin only' magnetic moment value of MO_4^{2-} is _____ BM. (Where M is a metal having least metallic radii. among Sc, Ti, V, Cr, Mn and Zn).

(Given atomic number: $Sc = 21, Ti = 22, V = 23, Cr = 24, Mn = 25$ and $Zn = 30$)

Open solution

Q25. JEE Main 2024 (Online) 6th April Morning Shift

The difference in the 'spin-only' magnetic moment values of $KMnO_4$ and the manganese product formed during titration of $KMnO_4$ against oxalic acid in acidic medium is _____ BM. (nearest integer)

Open solution

Q26. JEE Main 2024 (Online) 5th April Morning Shift

The spin-only magnetic moment value of the ion among Ti^{2+}, V^{2+}, Co^{3+} and Cr^{2+}, that acts as strong oxidising agent in aqueous solution is _____ BM (Near integer).

(Given atomic numbers : $Ti : 22, V : 23, Cr : 24, Co : 27$)

Open solution

Q27. JEE Main 2024 (Online) 31st January Morning Shift

The 'Spin only' Magnetic moment for $[Ni(NH_3)_6]^{2+}$ is _____ $\times 10^{-1}$ BM. (given = Atomic number of Ni : 28)

Open solution

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Q28. JEE Main 2024 (Online) 30th January Evening Shift

Number of complexes which show optical isomerism among the following is _____.

cis- $[Cr(ox)_2Cl_2]^{3-}$, $[Co(en)_3]^{3+}$, cis- $[Pt(en)_2Cl_2]^{2+}$, cis- $[Co(en)_2Cl_2]^{+}$, trans- $[Pt(en)_2Cl_2]^{2+}$, trans- $[Cr(ox)_2Cl_2]^{3-}$

Open solution

Q29. JEE Main 2024 (Online) 27th January Evening Shift

The Spin only magnetic moment value of square planar complex $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$ is _____ B.M. (Nearest integer)

(Given atomic number for $Pt = 78$)

Open solution

Q30. JEE Main 2023 (Online) 15th April Morning Shift

The homoleptic and octahedral complex of Co^{2+} and H_2O has _____ unpaired electron(s) in the t_{2g} set of orbitals.

Open solution

Q31. JEE Main 2023 (Online) 11th April Morning Shift

The ratio of spin-only magnetic moment values $\mu_{eff}\,[Cr(CN)_6]^{3-}/\mu_{eff}\,[Cr(H_2O)_6]^{3+}$ is _____.

Open solution

Q32. JEE Main 2023 (Online) 10th April Evening Shift

For a metal ion, the calculated magnetic moment is $4.90\ BM$. This metal ion has _____ number of unpaired electrons.

Open solution

Q33. JEE Main 2023 (Online) 10th April Morning Shift

In potassium ferrocyanide, there are _____ pairs of electrons in the t_{2g} set of orbitals.

Open solution

Q34. JEE Main 2023 (Online) 8th April Evening Shift

The observed magnetic moment of the complex $[Mn(\underline{N}CS)_6]^{x-}$ is $6.06\ BM$. The numerical value of x is _____.

Open solution

Q35. JEE Main 2023 (Online) 6th April Morning Shift

Number of ambidentate ligands in a representative metal complex $[M(en)(SCN)_4]$ is _____.

[en = ethylenediamine]

Open solution

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Q36. JEE Main 2023 (Online) 1st February Evening Shift

The spin only magnetic moment of $[Mn(H_2O)_6]^{2+}$ complexes is _____ B.M. (Nearest integer)

(Given : Atomic no. of Mn is 25)

Open solution

Q37. JEE Main 2023 (Online) 31st January Evening Shift

If the CFSE of $[Ti(H_2O)_6]^{3+}$ is $-96.0\,kJ/mol$, this complex will absorb maximum at wavelength _____ nm. (nearest integer)

Assume Planck's constant (h) $= 6.4 \times 10^{-34} Js$, Speed of light $(c) = 3.0 \times 10^8\ m/s$ and Avogadro's

Constant $(N_A) = 6 \times 10^{23}/mol$

Open solution

Q38. JEE Main 2023 (Online) 29th January Morning Shift

The sum of bridging carbonyls in $W(CO)_6$ and $Mn_2(CO)_{10}$ is _____.

Open solution

Q39. JEE Main 2023 (Online) 25th January Evening Shift

Total number of moles of AgCl precipitated on addition of excess of $AgNO_3$ to one mole each of the following complexes $[Co(NH_3)_4Cl_2]Cl$, $[Ni(H_2O)_6]Cl_2$, $[Pt(NH_3)_2Cl_2]$ and $[Pd(NH_3)_4]Cl_2$ is _____.

Open solution

Q40. JEE Main 2023 (Online) 25th January Morning Shift

The number of paramagnetic species from the following is _____.

$[Ni(CN)_4]^{2-}$, $[Ni(CO)_4]$, $[NiCl_4]^{2-}$

$[Fe(CN)_6]^{4-}$, $[Cu(NH_3)_4]^{2+}$

$[Fe(CN)_6]^{3-}$ and $[Fe(H_2O)_6]^{2+}$

Open solution

Q41. JEE Main 2023 (Online) 24th January Morning Shift

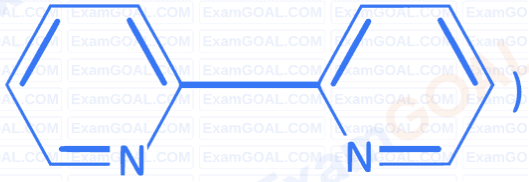
The d-electronic configuration of $[CoCl_4]^{2-}$ in tetrahedral crystal field is $e^m t_2^n$. Sum of "m" and "number of unpaired electrons" is _____

Open solution

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Q42. JEE Main 2022 (Online) 29th July Evening Shift

Sum of oxidation state (magnitude) and coordination number of cobalt in $Na\,[Co(bpy)Cl_4]$ is _____.

(Given : bpy = )

Open solution

Q43. JEE Main 2022 (Online) 29th July Morning Shift

$[Fe(CN)_6]^{3-}$ should be an inner orbital complex. Ignoring the pairing energy, the value of crystal field stabilization energy for this complex is $(-)$ _____ Δ_0. (Nearest integer)

Open solution

Q44. JEE Main 2022 (Online) 28th July Morning Shift

Total number of relatively more stable isomer(s) possible for octahedral complex $[Cu(en)_2(SCN)_2]$ will be _____.

Open solution

Q45. JEE Main 2022 (Online) 27th July Evening Shift

The spin only magnetic moment of the complex present in Fehling's reagent is _____ B.M. (Nearest integer).

Open solution

Q46. JEE Main 2022 (Online) 27th July Morning Shift

The conductivity of a solution of complex with formula $CoCl_3(NH_3)_4$ corresponds to 1 : 1 electrolyte, then the primary valency of central metal ion is _____.

Open solution

Q47. JEE Main 2022 (Online) 26th July Morning Shift

The difference between spin only magnetic moment values of $[Co(H_2O)_6]Cl_2$ and $[Cr(H_2O)_6]Cl_3$ is _____.

Open solution

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Q48. JEE Main 2022 (Online) 25th July Morning Shift

Consider the following metal complexes :

$[Co(NH_3)_6]^{3+}$

$[CoCl(NH_3)_5]^{2+}$

$[Co(CN)_6]^{3-}$

$[Co(NH_3)_5\,(H_2O)]^{3+}$

The spin-only magnetic moment value of the complex that absorbs light with shortest wavelength is _____ B. M. (Nearest integer)

Open solution

Q49. JEE Main 2022 (Online) 30th June Morning Shift

In the following brown complex, the oxidation state of iron is +_____.

$[Fe(H_2O)_6]^{2+} + NO \rightarrow \underset{\text{Brown complex}}{[Fe(H_2O)_5(NO)]^{2+}} + H_2O$

Open solution

Q50. JEE Main 2022 (Online) 30th June Morning Shift

Spin only magnetic moment (μ_s) of $K_3[Fe(CN)_6]$ is _____ B.M.

(Nearest integer)

Open solution

Q51. JEE Main 2022 (Online) 29th June Evening Shift

For the reaction given below :

$CoCl_3 \,.\, xNH_3 + AgNO_3\ (aq) \rightarrow$

If two equivalents of AgCl precipitate out, then the value of x will be _____.

Open solution

Q52. JEE Main 2022 (Online) 28th June Evening Shift

$[Fe(CN)_6]^{4-}$

$[Fe(CN)_6]^{3-}$

$[Ti(CN)_6]^{3-}$

$[Ni(CN)_4]^{2-}$

$[Co(CN)_6]^{3-}$

Among the given complexes, number of paramagnetic complexes is _____.

Open solution

Q53. JEE Main 2022 (Online) 28th June Evening Shift

(a) $CoCl_3.4NH_3$, (b) $CoCl_3.5NH_3$, (c) $CoCl_3.6NH_3$ and (d) $CoCl(NO_3)_2.5NH_3$.

Number of complex(es) which will exist in cis-trans form is/are _____.

Open solution

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Q54. JEE Main 2022 (Online) 28th June Morning Shift

Number of complexes which will exhibit synergic bonding amongst, $[Cr(CO)_6]$, $[Mn(CO)_5]$ and $[Mn_2(CO)_{10}]$ is _____.

Open solution

Q55. JEE Main 2022 (Online) 27th June Morning Shift

Acidified potassium permanganate solution oxidises oxalic acid. The spin-only magnetic moment of the manganese product formed from the above reaction is _____ B.M. (Nearest integer)

Open solution

Q56. JEE Main 2022 (Online) 26th June Evening Shift

Reaction of $[Co(H_2O)_6]^{2+}$ with excess ammonia and in the presence of oxygen results into a diamagnetic product. Number of electrons present in t_{2g}-orbitals of the product is _____.

Open solution

Q57. JEE Main 2022 (Online) 26th June Morning Shift

The spin-only magnetic moment value of an octahedral complex among $CoCl_3.4NH_3$, $NiCl_2.6H_2O$ and $PtCl_4.2HCl$, which upon reaction with excess of $AgNO_3$ gives 2 moles of AgCl is _____ B.M. (Nearest integer)

Open solution

Q58. JEE Main 2022 (Online) 25th June Evening Shift

Amongst $FeCl_3.3H_2O$, $K_3[Fe(CN)_6]$ and $[Co(NH_3)_6]Cl_3$, the spin-only magnetic moment value of the inner-orbital complex that absorbs light at shortest wavelength is _____ B.M. [nearest integer]

Open solution

Q59. JEE Main 2022 (Online) 25th June Morning Shift

If $[Cu(H_2O)_4]^{2+}$ absorbs a light of wavelength 600 nm for d-d transition, then the value of octahedral crystal field splitting energy for $[Cu(H_2O)_6]^{2+}$ will be _____ $\times 10^{-21}$ J. [Nearest Integer]

(Given : h = 6.63 $\times 10^{-34}$ Js and c = 3.08 $\times 10^{8}$ ms^{-1})

Open solution

Q60. JEE Main 2022 (Online) 24th June Morning Shift

In the cobalt-carbonyl complex : $[Co_2(CO)_8]$, number of Co-Co bonds is "X" and terminal CO ligands is "Y". X + Y = _____.

Open solution

Q61. JEE Main 2021 (Online) 1st September Evening Shift

The sum of oxidation states of two silver ions in $[Ag(NH_3)_2]$ $[Ag(CN)_2]$ complex is _____.

Open solution

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Q62. JEE Main 2021 (Online) 27th August Evening Shift

The number of optical isomers possible for $[Cr(C_2O_4)_3]^{3-}$ is _____.

Open solution

Q63. JEE Main 2021 (Online) 27th August Morning Shift

1 mol of an octahedral metal complex with formula MCl_3 . 2L on reaction with excess of $AgNO_3$ gives 1 mol of AgCl. The denticity of Ligand L is _____. (Integer answer)

Open solution

Q64. JEE Main 2021 (Online) 26th August Evening Shift

The overall stability constant of the complex ion $[Cu(NH_3)_4]^{2+}$ is 2.1×10^{13}. The overall dissociations constant is $y \times 10^{-14}$. Then y is _____. (Nearest integer)

Open solution

Q65. JEE Main 2021 (Online) 26th August Morning Shift

The ratio of number of water molecules in Mohr's salt and potash alum is _____ $\times 10^{-1}$. (Integer answer)

Open solution

Q66. JEE Main 2021 (Online) 27th July Evening Shift

3 moles of metal complex with formula $Co(en)_2Cl_3$ gives 3 moles of silver chloride on treatment with excess of silver nitrate. The secondary valency of Co in the complex is _____.

(Round off to the nearest integer)

Open solution

Q67. JEE Main 2021 (Online) 27th July Morning Shift

The number of geometrical isomers possible in triamminetrinitrocobalt (III) is X and in trioxalatochromate (III) is Y. Then the value of X + Y is _____.

Open solution

Q68. JEE Main 2021 (Online) 25th July Morning Shift

Three moles of AgCl get precipitated when one mole of an octahedral co-ordination compound with empirical formula $CrCl_3.3NH_3.3H_2O$ reacts with excess of silver nitrate. The number of chloride ions satisfying the secondary valency of the metal ion is _____.

Open solution

Q69. JEE Main 2021 (Online) 22th July Evening Shift

The total number of unpaired electrons present in $[Co(NH_3)_6]Cl_2$ and $[Co(NH_3)_6]Cl_3$ is :

Open solution

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Q70. JEE Main 2021 (Online) 20th July Evening Shift

An aqueous solution of $NiCl_2$ was heated with excess sodium cyanide in presence of strong oxidizing agent to form $[Ni(CN)_6]^{2-}$. The total change in number of unpaired electrons on metal centre is _____.

Open solution

Q71. JEE Main 2021 (Online) 20th July Morning Shift

The spin-only magnetic moment value for the complex $[Co(CN)_6]^{4-}$ is _____ BM.

[At. no. of Co = 27]

Open solution

Q72. JEE Main 2021 (Online) 18th March Morning Shift

The total number of unpaired electrons present in the complex $K_3[Cr(oxalate)_3]$ is _____.

Open solution

Q73. JEE Main 2021 (Online) 17th March Evening Shift

On complete reaction of $FeCl_3$ with oxalic acid in aqueous solution containing KOH, resulted in the formation of product A. The secondary valency of Fe in the product A is _____. (Round off to the Nearest Integer).

Open solution

Q74. JEE Main 2021 (Online) 16th March Evening Shift

$[Ti(H_2O)_6]^{3+}$ absorbs light of wavelength 498 nm during a d $-$ d transition. The octahedral splitting energy for the above complex is _____ $\times 10^{-19}$ J. (Round off to the Nearest Integer). h = 6.626×10^{-34} Js; c = 3×10^8 ms^{-1}

Open solution

Q75. JEE Main 2021 (Online) 16th March Morning Shift

The equivalents of ethylene diamine required to replace the neutral ligands from the coordination sphere of the trans-complex of $CoCl_3 \cdot 4NH_3$ is _____. (Round off to the Nearest Integer).

Open solution

Q76. JEE Main 2021 (Online) 26th February Evening Shift

The number of stereoisomers possible for $[Co(ox)_2(Br)(NH_3)]^{2-}$ is _____. [ox = oxalate]

Open solution

Q77. JEE Main 2021 (Online) 26th February Morning Shift

Number of bridging CO ligands in $[Mn_2(CO)_{10}]$ is _____.

Open solution

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Q78. JEE Main 2021 (Online) 25th February Evening Shift

The spin only magnetic moment of a divalent ion in aqueous solution (atomic number 29) is _____ BM.

Open solution

Q79. JEE Main 2020 (Online) 5th September Evening Slot

Considering that $\Delta_0 > P$, the magnetic moment (in BM) of $[Ru(H_2O)_6]^{2+}$ would be _____.

Open solution

Q80. JEE Main 2020 (Online) 5th September Morning Slot

The total number of coordination sites in ethylenediaminetetraacetate ($EDTA^{4-}$) is ____.

Open solution

Q81. JEE Main 2020 (Online) 2nd September Morning Slot

The oxidation states of iron atoms in compounds (A), (B) and (C), respectively, are x, y and z. The sum of x, y and z is _____.

$Na_4[Fe(CN)_5(NOS)]$
(A)

$Na_4[FeO_4]$
(B)

$[Fe_2(CO)_9]$
(C)

Open solution

Q82. JEE Main 2020 (Online) 8th January Evening Slot

Complexes (ML_5) of metals Ni and Fe have ideal square pyramidal and trigonal bipyramidal geometries, respectively. The sum of the 90°, 120° and 180° L-M-L angles in the two complexes is _____.

Open solution

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MCQ (Single Correct Answer) 304 questions

Q83. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Which of the following sequences of hybridisation, geometry and magnetic nature are correct for the given coordination compounds?

A. $[NiCl_4]^{2-}$ – sp^3, tetrahedral, paramagnetic

B. $[Ni(NH_3)_6]^{2+}$ – $sp^3\,d^2$, octahedral, paramagnetic

C. $[Ni(CO)_4]$ – sp^3, tetrahedral, paramagnetic

D. $[Ni(CN)_4]^{2-}$ – dsp^2, square planar, diamagnetic

Choose the correct answer from the options given below :

(A) A, B, C and D **(B)** B, C and D only **(C)** A, C and D only **(D)** A, B and D only

Q84. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

Match the LIST-I with LIST-II

List-I Electronic configuration of tetrahedral metal ion		**List-II Crystal Field Stabilization Energy (Δ_t)**	
A.	d^2	I.	-0.6
B.	d^4	II.	-0.8
C.	d^6	III.	-1.2
D.	d^8	IV.	-0.4

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-II, D-I **(B)** A-III, B-I, C-IV, D-II

(C) A-III, B-IV, C-I, D-II **(D)** A-II, B-I, C-IV, D-III

Q85. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

Which of the following are true about the energy of the given d-orbitals of a tetrahedral complex?

A. $d_{xy} = d_{xz} > d_{x^2-y^2}$

B. $d_{xy} = d_{yz} > d_{z^2}$

C. $d_{x^2-y^2} > d_{z^2} > d_{xz}$

D. $d_{x^2-y^2} = d_{z^2} < d_{xz}$

Choose the correct answer from the given below :

(A) A, B and D only **(B)** A and B only **(C)** B and D only **(D)** B, C and D only

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Q86. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Given below are two statements :

Statement I : Presence of large number of unpaired electrons in transition metal atoms results in higher enthalpies of their atomisation.

Statement II : $d_{xy} = d_{xz} = d_{yz} < d_{x^2-y^2} = d_{z^2}$ and $d_{x^2-y^2} = d_{z^2} < d_{xy} = d_{xz} = d_{yz}$ are the d-orbital splittings in $[Fe(H_2O)_6]^{3+}$ and $[Ni(Cl)_4]^{2-}$ complex ions respectively.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q87. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Identify the correct statements from the following

A. $[Fe(C_2O_4)_3]^{3-}$ is the most stable complex among $[Fe(OH)_6]^{3-}$, $[Fe(C_2O_4)_3]^{3-}$ and $[Fe(SCN)_6]^{3-}$

B. The stability of $[Cu(NH_3)_4]^{2+}$ is greater than that of $[Cu(en)_2]^{2+}$

C. The hybridization of Fe in $K_4[Fe(CN)_6]$ is d^2sp^3

D. $[Fe(NO_2)_3Cl_3]^{3-}$ exhibits linkage isomerism

E. NO_2^- and SCN^- ligands are NOT ambidentate ligands

Choose the correct answer from the options given below :

(A) A, B, C, D and E (B) B, C and D only (C) A, C and D only (D) A, C and E only

Q88. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

The correct statements about metal carbonyls are

A. The metal-carbon bonds in metal carbonyls possess both σ and π character.

B. Due to synergic bonding interactions between metal and CO ligand, the metal-carbon bond becomes weak.

C. The metal-carbon σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of metal.

D. The metal-carbon π bond is formed by the donation of electrons from filled d-orbital of metal into vacant π^* orbital of CO .

Choose the correct answer from the options given below :

(A) A and B Only (B) A, C and D Only (C) B and C Only (D) A and D Only

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Q89. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Given below are two statements:

Statement I: Each electron in e_g orbitals destabilizes the orbitals by $+0.6\Delta_o$ and each electron in the t_{2g} orbitals stabilizes the orbitals by $-0.4\Delta_0$ in an octahedral field on the basis of crystal field theory.

Statement II: All the d - orbitals of the transition metals have the same energy in their free atomic state but when a complex is formed the ligands destroy the degeneracy of these orbitals on the basis of crystal field theory.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q90. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

Consider the metal complexes $[Ni(en)_3]^{2+}(A)$, $[NiCl_4]^{2-}(B)$ and $[Ni(NH_3)_6]^{2+}(C)$. Choose the CORRECT option by considering the number of unpaired electrons present in (A), (B) and (C) respectively and the order of frequency of absorption.

(A) 2, 2, 2 and $(A) > (C) > (B)$

(B) 0, 2, 0 and $(A) > (C) > (B)$

(C) 2, 2, 0 and $(B) > (C) > (A)$

(D) 2, 2, 2 and $(C) > (A) > (B)$

Q91. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

Match the LIST-I with LIST-II

List-I Complex ion		**List-II Calculated spin only magnetic moment (BM)**	
A.	$[Cr(H_2O)_6]^{2+}$	I.	3.87
B.	$[Co(H_2O)_6]^{2+}$	II.	5.92
C.	$[Cu(H_2O)_6]^{2+}$	III.	4.90
D.	$[Mn(H_2O)_6]^{2+}$	IV.	1.73

Choose the correct answer from the options given below :

(A) A-I, B-III, C-IV, D-II

(B) A-II, B-I, C-III, D-IV

(C) A-IV, B-II, C-I, D-III

(D) A-III, B-I, C-IV, D-II

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Q92. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Given below are two statements :

Statement I : Among Zn, Mn, Sc and Cu, the energy required to remove the third valence electron is highest for Zn and lowest for Sc.

Statement II : The correct order of the following complexes in terms of CFSE is $[Co(H_2O)_6]^{2+} < [Co(H_2O)_6]^{3+} < [Co(en)_3]^{3+}$.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both **Statement I** and **Statement II** are true

(B) Both **Statement I** and **Statement II** are false

(C) **Statement I** is true but **Statement II** is false

(D) **Statement I** is false but **Statement II** is true

Q93. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Which of the following complexes will show coordination isomerism?

A. $[Ag(NH_3)_2][Ag(CN)_2]$

B. $[Co(NH_3)_6][Cr(CN)_6]$

C. $[Co(NH_3)_6][Co(CN)_6]$

D. $[Fe(NH_3)_6][Co(CN)_6]$

E. $[Co(NH_3)_6][Fe(CN)_6]$

Choose the **correct** answer from the options given below:

(A) B, C and D Only **(B)** B, D and E Only **(C)** A, C and D Only **(D)** C, D and E Only

Q94. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Match List - I with List - II.

	List - I **Chromium (III) Complexes (en = ethylene diamine)**	**List - II** **Δ_o (cm^{-1})**
A	$[Cr(CN)_6]^{3-}$	I. 15,060
B	$[CrF_6]^{3-}$	II. 17,400
C	$[Cr(H_2O)_6]^{3+}$	III. 22,300
D	$[Cr(en)_3]^{3+}$	IV. 26,600

Choose the **correct** answer from the options given below:

(A) A-I, B-II, C-III, D-IV **(B)** A-II, B-III, C-IV, D-I

(C) A-III, B-IV, C-I, D-II **(D)** A-IV, B-I, C-II, D-III

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Q95. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

The correct increasing order of spin-only magnetic moment values of the complex ions

$[MnBr_4]^{2-}$ (A), $[Cu(H_2O)_6]^{2+}$ (B), $[Ni(CN)_4]^{2-}$ (C) and $[Ni(H_2O)_6]^{2+}$ (D) is :

(A) A = B < D < C (B) C = D < B < A (C) C < B < D < A (D) A = B < C < D

Q96. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

The correct statement among the following is:

(A) $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are diamagnetic and $Ni(CO)_4$ is paramagnetic.

(B) $Ni(CO)_4$ is diamagnetic and $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ are paramagnetic.

(C) $Ni(CO)_4$ and $[NiCl_4]^{2-}$ are diamagnetic and $[Ni(CN)_4]^{2-}$ is paramagnetic.

(D) $Ni(CO)_4$ and $[Ni(CN)_4]^{2-}$ are diamagnetic and $[NiCl_4]^{2-}$ is paramagnetic.

Q97. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

The wavelength of light absorbed for the following complexes are in the order

$[Co(NH_3)_6]^{3+}$; $[Co(H_2O)_6]^{3+}$; $[Co(CN)_6]^{3-}$; $[Co(NH_3)_5(H_2O)]^{3+}$; $[CoF_6]^{3-}$

(I) (II) (III) (IV) (V)

(A) III < IV < I < II < V (B) III < I < IV < V < II

(C) III < I < IV < II < V (D) III < I < II < IV < V

Q98. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Given below are two statements :

Statement I : Hybridisation, shape and spin only magnetic moment of $K_3[Co(CO_3)_3]$ is sp^3d^2, octahedral and 4.9 BM respectively.

Statement II : Geometry, hybridisation and spin only magnetic moment values (BM) of the ions $[Ni(CN)_4]^{2-}$, $[MnBr_4]^{2-}$ and $[CoF_6]^{3-}$ respectively are square planar, tetrahedral, octahedral; dsp^2, sp^3, sp^3d^2 and $0, 5.9, 4.9$.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

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Q99. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Given below are two statements:

Statement I : The number of paramagnetic species among $[CoF_6]^{3-}$, $[TiF_6]^{3-}$, V_2O_5 and $[Fe(CN)_6]^{3-}$ is 3 .

Statement II :

$K_4[Fe(CN)_6] < K_3[Fe(CN)_6] < [Fe(H_2O)_6]SO_4 \cdot H_2O < [Fe(H_2O)_6]Cl_3$ is the correct order in terms of number of unpaired electron(s) present in the complexes.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

Q100. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Consider a mixture ' X ' which is made by dissolving 0.4 mol of $[Co(NH_3)_5SO_4]Br$ and 0.4 mol of $[Co(NH_3)_5Br]SO_4$ in water to make 4 L of solution. When 2 L of mixture ' X ' is allowed to react with excess of $AgNO_3$, it forms precipitate ' Y '. The rest 2 L of mixture ' X ' reacts with excess $BaCl_2$ to form precipitate ' Z '. Which of the following statements is CORRECT?

(A) 0.1 mol of ' Y ' is formed.

(B) 0.4 mol of ' Z ' is formed.

(C) 0.2 mol of ' Z ' is formed.

(D) ' Y ' is $BaSO_4$ and ' Z ' is AgBr .

Q101. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

Identify the CORRECT set of details from the following :

A. $[Co(NH_3)_6]^{3+}$: Inner orbital complex; d^2sp^3 hybridized

B. $[MnCl_6]^{3-}$: Outer orbital complex; sp^3d^2 hybridized

C. $[CoF_6]^{3-}$: Outer orbital complex; d^2sp^3 hybridized

D. $[FeF_6]^{3-}$: Outer orbital complex; sp^3d^2 hybridized

E. $[Ni(CN)_4]^{2-}$: Inner orbital complex; sp^3 hybridized

Choose the correct answer from the options given below :

(A) C & D Only

(B) A, C & E Only

(C) A, B, C, D & E

(D) A, B & D Only

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Q102. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

The statements that are incorrect about the nickel(II) complex of dimethylglyoxime are :

A. It is red in colour.

B. It has a high solubility in water at $\mathrm{pH} = 9$.

C. The Ni ion has two unpaired d-electrons.

D. The $\mathrm{N} - \mathrm{Ni} - \mathrm{N}$ bond angle is almost close to 90°.

E. The complex contains four five-membered metallacycles (metal containing rings).

Choose the correct answer from the options given below :

(A) C and E Only **(B)** A, D and B Only **(C)** B, C and E Only **(D)** C and D Only

Q103. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Given below are two statements :

Statement I : $[\mathrm{CoBr_4}]^{2-}$ ion will absorb light of lower energy than $[\mathrm{CoCl_4}]^{2-}$ ion.

Statement II : In $[\mathrm{CoI_4}]^{2-}$ ion, the energy separation between the two set of d-orbitals is more than $[\mathrm{CoCl_4}]^{2-}$ ion.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q104. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

$[\mathrm{Ni(PPh_3)_2Cl_2}]$ is a paramagnetic complex. Identify the INCORRECT statements about this complex.

A. The complex exhibits geometrical isomerism.

B. The complex is white in colour.

C. The calculated spin-only magnetic moment of the complex is 2.84 BM .

D. The calculated CFSE (Crystal Field Stabilization Energy) of Ni in this complex is $-0.8\Delta_o$.

E. The geometrical arrangement of ligands in this complex is similar to that in $\mathrm{Ni(CO)_4}$.

Choose the correct answer from the options given below :

(A) C and D Only **(B)** A, B and D Only **(C)** C, D and E Only **(D)** A and B Only

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Q105. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Consider the transition metal ions $Mn^{3+}, Cr^{3+}, Fe^{3+}$ and Co^{3+} and all form low spin octahedral complexes. The correct decreasing order of unpaired electrons in their respective d-orbitals of the complexes is

(A) $Cr^{3+} > Mn^{3+} > Fe^{3+} > Co^{3+}$

(B) $Mn^{3+} > Fe^{3+} > Co^{3+} > Cr^{3+}$

(C) $Fe^{3+} > Co^{3+} > Mn^{3+} > Cr^{3+}$

(D) $Cr^{3+} > Fe^{3+} > Co^{3+} > Mn^{3+}$

Q106. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

A first row transition metal (M) does not liberate H_2 gas from dilute HCl .1 mol of aqueous solution of MSO_4 is treated with excess of aqueous KCN and then $H_2S(g)$ is passed through the solution. The amount of MS (metal sulphide) formed from the above reaction is ____ mol.

(A) 1 (B) 2 (C) 0 (D) 3

Q107. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Given below are two statements :

Statement I : Crystal Field Stabilization Energy (CFSE) of $[Cr(H_2O)_6]^{2+}$ is greater than that of $[Mn(H_2O)_6]^{2+}$.

Statement II : Potassium ferricyanide has a greater spin-only magnetic moment than sodium ferrocyanide.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) **Statement I** is true but **Statement II** is false

(B) Both **Statement I** and **Statement II** are true

(C) Both **Statement I** and **Statement II** are false

(D) **Statement I** is false but **Statement II** is true

Q108. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Given below are two statements :

Statement I : Among $[Cu(NH_3)_4]^{2+}, [Ni(en)_3]^{2+}, [Ni(NH_3)_6]^{2+}$ and $[Mn(H_2O)_6]^{2+}, [Mn(H_2O)_6]^{2+}$ has the maximum number of unpaired electrons.

Statement II : The number of pairs among $\left\{[NiCl_4]^{2-}, [Ni(CO)_4]\right\}, \left\{[NiCl_4]^{2-}, [Ni(CN)_4]^{2-}\right\}$ and $\left\{[Ni(CO)_4], [Ni(CN)_4]^{2-}\right\}$ that contain only diamagnetic species is two.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

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Q109. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Match the **LIST-I** with **LIST-II**

LIST-I (Complex/ Species)	LIST-II (Shape & magnetic moment)
A. $[Ni(CO)_4]$	I. Tetrahedral, 2.8 BM
B. $[Ni(CN)_4]^{2-}$	II. Square planar, 0 BM
C. $[NiCl_4]^{2-}$	III. Tetrahedral, 0 BM
D. $[MnBr_4]^{2-}$	IV. Tetrahedral, 5.9 BM

Choose the **correct** answer from the options given below:

(A) A-I, B-II, C-III, D-IV

(B) A-IV, B-I, C-III, D-II

(C) A-III, B-II, C-I, D-IV

(D) A-III, B-IV, C-II, D-I

Q110. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

The number of species from the following that are involved in sp^3d^2 hybridization is :

$[Co(NH_3)_6]^{3+}$, SF_6, $[CrF_6]^{3-}$, $[CoF_6]^{3-}$, $[Mn(CN)_6]^{3-}$, and $[MnCl_6]^{3-}$

(A) 4 **(B)** 3 **(C)** 5 **(D)** 6

Q111. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Given below are two statements:

Statement I: A homoleptic octahedral complex, formed using monodentate ligands, will not show stereoisomerism.

Statement II: cis- and trans- platin are heteroleptic complexes of Pd.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

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Q112. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Match **List - I** with **List - II**.

List - I (Complex)	List - II (Primary valency and Secondary valency)
(A) $[Co(en)_2Cl_2]Cl$	(I) 3, 6
(B) $[Pt(NH_3)_2Cl(NO_2)]$	(II) 3, 4
(C) Hg $[Co(SCN)_4]$	(III) 2, 6
(D) $[Mg\ (EDTA)]^{2-}$	(IV) 2, 4

Choose the **correct** answer from the options given below :

(A) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

(B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(C) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

(D) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Q113. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

The number of unpaired electrons responsible for the paramagnetic nature of the following complex species are respectively :

$[Fe(CN)_6]^{3-}$, $[FeF_6]^{3-}$, $[CoF_6]^{3-}$, $[Mn(CN)_6]^{3-}$

(A) 1, 5, 4, 2 **(B)** 1, 4, 4, 2 **(C)** 1, 5, 5, 2 **(D)** 1, 1, 4, 2

Q114. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

'X' is the number of acidic oxides among VO_2, V_2O_3, CrO_3, V_2O_5 and Mn_2O_7. The primary valency of cobalt in
$[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$ is Y. The value of X + Y is _____.

(A) 3 **(B)** 4 **(C)** 2 **(D)** 5

Q115. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

An octahedral complex having molecular composition $Co \cdot 5NH_3 \cdot Cl \cdot SO_4$ has two isomers A and B. The solution of A gives a white precipitate with $AgNO_3$ solution and the solution of B gives white precipitate with $BaCl_2$ solution. The type of isomerism exhibited by the complex is,

(A) Co-ordinate isomerism

(B) Ionisation isomerism

(C) Geometrical isomerism

(D) Linkage isomerism

Q116. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

' X ' is the number of electrons in t_{2g} orbitals of the most stable complex ion among $[Fe(NH_3)_6]^{3+}$, $[FeCl_6]^{3-}$, $[Fe(C_2O_4)_3]^{3-}$ and $[Fe(H_2O)_6]^{3+}$. The nature of oxide of vanadium of the type V_2O_X is :

(A) Amphoteric **(B)** Acidic **(C)** Basic **(D)** Neutral

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Q117. JEE Main 2025 (Online) 4th April Evening Shift

The correct order of $[FeF_6]^{3-}$, $[CoF_6]^{3-}$, $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ complex species based on the number of unpaired electrons present is:

(A) $[CoF_6]^{3-} > [FeF_6]^{3-} > [Ni(CO)_4] > [Ni(CN)_4]^{2-}$

(B) $[FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CN)_4]^{2-} > [Ni(CO)_4]$

(C) $[FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CN)_4]^{2-} = [Ni(CO)_4]$

(D) $[Ni(CN)_4]^{2-} > [FeF_6]^{3-} > [CoF_6]^{3-} > [Ni(CO)_4]$

Open solution

Q118. JEE Main 2025 (Online) 4th April Morning Shift

Which one of the following complexes will have $\Delta_o = 0$ and $\mu = 5.96$ B.M?

(A) $[FeF_6]^{4-}$

(B) $[Mn(SCN)_6]^{4-}$

(C) $[Co(NH_3)_6]^{3+}$

(D) $[Fe(CN)_6]^{4-}$

Open solution

Q119. JEE Main 2025 (Online) 4th April Morning Shift

Number of stereoisomers possible for the complexes, $[CrCl_3(py)_3]$ and $[CrCl_2(ox)_2]^{3-}$ are respectively (py = pyridine, ox = oxalate)

(A) 3 & 3

(B) 2 & 3

(C) 1 & 2

(D) 2 & 2

Open solution

Q120. JEE Main 2025 (Online) 3rd April Evening Shift

Identify the diamagnetic octahedral complex ions from below ;

A. $[Mn(CN)_6]^{3-}$

B. $[Co(NH_3)_6]^{3+}$

C. $[Fe(CN)_6]^{4-}$

D. $[Co(H_2O)_3F_3]$

Choose the correct answer from the options given below:

(A) B and C Only

(B) A and C Only

(C) B and D Only

(D) A and D Only

Open solution

Q121. JEE Main 2025 (Online) 3rd April Morning Shift

The correct order of the complexes $[Co(NH_3)_5(H_2O)]^{3+}(A)$, $[Co(NH_3)_6]^{3+}(B)$, $[Co(CN)_6]^{3-}(C)$ and $[CoCl(NH_3)_5]^{2+}(D)$ in terms of wavelength of light absorbed is

(A) $D > C > B > A$

(B) $C > B > A > D$

(C) $D > A > B > C$

(D) $C > B > D > A$

Open solution

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Q122. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Match the LIST-I with LIST-II

LIST-I (Molecules/ion)		LIST-II (Hybridisation of central atom)	
A.	PF_5	I	dsp^2
B	SF_6	II	$sp^3\,d$
C	$Ni(CO)_4$	III	$sp^3\,d^2$
D	$[PtCl_4]^{2-}$	IV	sp^3

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-III, B-I, C-IV, D-II

(C) A-II, B-III, C-IV, D-I

(D) A-I, B-II, C-III, D-IV

Q123. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

The type of hybridization and the magnetic property of $[MnCl_6]^{3-}$ are,

(A) sp^3d^2, paramagnetic with four unpaired electrons.

(B) d^2sp^3, paramagnetic with four unpaired electrons.

(C) $sp^3\,d^2$, paramagnetic with two unpaired electrons.

(D) d^2sp^3, paramagnetic with two unpaired electrons.

Q124. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

The d-orbital electronic configuration of the complex among $[Co(en)_3]^{3+}$, $[CoF_6]^{3-}$, $[Mn(H_2O)_6]^{2+}$ and $[Zn(H_2O)_6]^{2+}$ that has the highest CFSE is :

(A) $t_{2g}^6e_g^0$

(B) $t_{2g}^3e_g^2$

(C) $t_{2g}^4e_g^2$

(D) $t_{2g}^6e_g^4$

Q125. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

Given below are two statements :

Statement (I) : In octahedral complexes, when $\Delta_0 < P$ high spin complexes are formed. When $\Delta_0 > P$ low spin complexes are formed.

Statement (II) : In tetrahedral complexes because of $\Delta_t < P$, low spin complexes are rarely formed.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are incorrect

(B) Statement I is correct but Statement II is incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

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Q126. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Identify the homoleptic complexes with odd number of d electrons in the central metal :

(A) $[FeO_4]^{2-}$

(B) $[Fe(CN)_6]^{3-}$

(C) $[Fe(CN)_5NO]^{2-}$

(D) $[CoCl_4]^{2-}$

(E) $[Co(H_2O)_3F_3]$

Choose the **correct** answer from the options given below :

(A) (C) and (E) only

(B) (A), (B) and (D) only

(C) (B) and (D) only

(D) (A), (C) and (E) only

Q127. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

The calculated spin-only magnetic moments of $K_3[Fe(OH)_6]$ and $K_4[Fe(OH)_6]$ respectively are :

(A) 4.90 and 4.90 B.M.

(B) 4.90 and 5.92 B.M.

(C) 5.92 and 4.90 B.M.

(D) 3.87 and 4.90 B.M.

Q128. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

The correct increasing order of stability of the complexes based on Δ_0 value is :

I. I. $[Mn(CN)_6]^{3-}$

II. II. $[Co(CN)_6]^{4-}$

III. III. $[Fe(CN)_6]^{4-}$

IV. IV. $[Fe(CN)_6]^{3-}$

(A) I < II < IV < III

(B) IV < III < II < I

(C) III < II < IV < I

(D) II < III < I < IV

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Q129. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

Match List - I with List - II.

List - I (Complex)	List - II (Hybridisation & Magnetic characters)
(A) $[MnBr_4]^{2-}$	(I) d^2sp^3 & diamagnetic
(B) $[FeF_6]^{3-}$	(II) sp^3d^2 & paramagnetic
(C) $[Co(C_2O_4)_3]^{3-}$	(III) sp^3 & diamagnetic
(D) $[Ni(CO)_4]$	(IV) sp^3 & paramagnetic

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(I), (D)-(IV)

(B) (A)-(III), (B)-(II), (C)-(III), (D)-(IV)

(C) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(D) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Q130. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Match List - I with List - II.

List - I (Complex)	List - II (Hybridisation of central metal ion)
(A) $[CoF_6]^{3-}$	(I) d^2sp^3
(B) $[NiCl_4]^{2-}$	(II) sp^3
(C) $[Co(NH_3)_6]^{3+}$	(III) sp^3d^2
(D) $[Ni(CN)_4]^{2-}$	(IV) dsp^2

Choose the correct answer from the options given below:

(A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(B) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(C) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

(D) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Q131. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

The conditions and consequence that favours the $t_{2g}^3 e_g^1$ configuration in a metal complex are :

(A) strong field ligand, low spin complex

(B) strong field ligand, high spin complex

(C) weak field ligand, high spin complex

(D) weak field ligand, low spin complex

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Q132. JEE Main 2025 (Online) 24th January Evening Shift

When Ethane-1,2-diamine is added progressively to an aqueous solution of Nickel (II) chloride, the sequence of colour change observed will be:

Open solution

(A) Pale Blue $\rightarrow$ Blue $\rightarrow$ Green $\rightarrow$ Violet

(B) Pale Blue $\rightarrow$ Blue $\rightarrow$ Violet $\rightarrow$ Green

(C) Green $\rightarrow$ Pale Blue $\rightarrow$ Blue $\rightarrow$ Violet

(D) Violet $\rightarrow$ Blue $\rightarrow$ Pale Blue $\rightarrow$ Green

Q133. JEE Main 2025 (Online) 24th January Morning Shift

One mole of the octahedral complex compound $Co(NH_3)_5Cl_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with excess of $AgNO_3$ solution to yield two moles of $AgCl_{(s)}$. The structure of the complex is:

Open solution

(A) $[Co(NH_3)_5Cl]Cl_2$

(B) $[Co(NH_3)_4Cl] \cdot Cl_2 \cdot NH_3$

(C) $[Co(NH_3)_3Cl_3] \cdot 2NH_3$

(D) $[Co(NH_3)_4Cl_2] \cdot Cl \cdot NH_3$

Q134. JEE Main 2025 (Online) 23rd January Evening Shift

Identify the coordination complexes in which the central metal ion has d^4 configuration.

Open solution

(A) $[FeO_4]^{2-}$

(B) $[Mn(CN)_6]^{3-}$

(C) $[Fe(CN)_6]^{3-}$

(D)



(E) $[NiF_6]^{2-}$

Choose the correct answer from the options given below :

(A) (B), (C) and (D) only

(B) (C) and (E) only

(C) $(A), (B)$ and (E) only

(D) (B) and (D) only

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Q135. JEE Main 2025 (Online) 23rd January Morning Shift

$CrCl_3 \cdot xNH_3$ can exist as a complex. 0.1 molal aqueous solution of this complex shows a depression in freezing point of $0.558^\circ C$. Assuming 100% ionisation of this complex and coordination number of Cr is 6 , the complex will be (Given $K_f = 1.86\ K\ kg\ mol^{-1}$)

Open solution

(A) $[Cr(NH_3)_5Cl]Cl_2$

(B) $[Cr(NH_3)_4Cl_2]Cl$

(C) $[Cr(NH_3)_6]Cl_3$

(D) $[Cr(NH_3)_3Cl_3]$

Q136. JEE Main 2025 (Online) 23rd January Morning Shift

The d-electronic configuration of an octahedral Co (II) complex having magnetic moment of 3.95 BM is:

Open solution

(A) $e^4t_2^3$

(B) $t_{2g}{}^6e_g{}^1$

(C) $t_{2g}{}^5e_g{}^2$

(D) $t_{2g}^3e_g^0$

Q137. JEE Main 2025 (Online) 23rd January Morning Shift

The complex that shows Facial - Meridional isomerism is :

Open solution

(A) $[Co(en)_2Cl_2]^+$

(B) $[Co(NH_3)_3Cl_3]$

(C) $[Co(NH_3)_4Cl_2]^+$

(D) $[Co(en)_3]^{3+}$

Q138. JEE Main 2025 (Online) 22nd January Evening Shift

The correct order of the following complexes in terms of their crystal field stabilization energies is :

Open solution

(A) $[Co(NH_3)_4]^{2+} < [Co(NH_3)_6]^{2+} < [Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+}$

(B) $[Co(NH_3)_4]^{2+} < [Co(NH_3)_6]^{2+} < [Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+}$

(C) $[Co(NH_3)_6]^{2+} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_4]^{2+} < [Co(en)_3]^{3+}$

(D) $[Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_6]^{2+} < [Co(NH_3)_4]^{2+}$

Q139. JEE Main 2025 (Online) 22nd January Evening Shift

Identify the homoleptic complex(es) that is/are low spin.

Open solution

(A) $[Fe(CN)_5NO]^{2-}$

(B) $[CoF_6]^{3-}$

(C) $[Fe(CN)_6]^{4-}$

(D) $[Co(NH_3)_6]^{3+}$

(E) $[Cr(H_2O)_6]^{2+}$

Choose the correct answer from the options given below :

(A) (C) and (D) only

(B) (C) only

(C) (A) and (C) only

(D) (B) and (E) only

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Q140. JEE Main 2025 (Online) 22nd January Morning Shift

From the magnetic behaviour of $[NiCl_4]^{2-}$ (paramagnetic) and $[Ni(CO)_4]$ (diamagnetic), choose the correct geometry and oxidation state.

Open solution

(A) $[NiCl_4]^{2-}$: $Ni(0)$, tetrahedral $[Ni(CO)_4]$: $Ni(0)$, square planar

(B) $[NiCl_4]^{2-}$: Ni^{II}, tetrahedral $[Ni(CO)_4]$: $Ni(0)$, tetrahedral

(C) $[NiCl_4]^{2-}$: Ni^{II}, tetrahedral $[Ni(CO)_4]$: Ni^{II}, square planar

(D) $[NiCl_4]^{2-}$: Ni^{II}, square planar $[Ni(CO)_4]$: $Ni(0)$, square planar

Q141. JEE Main 2025 (Online) 22nd January Morning Shift

In which of the following complexes the CFSE, Δ_o will be equal to zero?

Open solution

(A) $K_3[Fe(SCN)_6]$

(B) $[Fe(en)_3]Cl_3$

(C) $[Fe(NH_3)_6]Br_2$

(D) $K_4[Fe(CN)_6]$

Q142. JEE Main 2024 (Online) 9th April Evening Shift

Match List I with List II

Open solution

	LIST I		LIST II
A.	$K_2[Ni(CN)_4]$	I.	sp^3
B.	$[Ni(CO)_4]$	II.	sp^3d^2
C.	$[Co(NH_3)_6]Cl_3$	III.	dsp^2
D.	$Na_3[CoF_6]$	IV.	d^2sp^3

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I

(B) A-III, B-I, C-II, D-IV

(C) A-I, B-III, C-II, D-IV

(D) A-III, B-I, C-IV, D-II

Q143. JEE Main 2024 (Online) 9th April Evening Shift

The coordination environment of Ca^{2+} ion in its complex with $EDTA^{4-}$ is :

Open solution

(A) trigonal prismatic

(B) octahedral

(C) square planar

(D) tetrahedral

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Q144. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The total number of geometrical isomers shown by $[Co(en)_2Cl_2]^+$ complex ion is three.

Reason (R): $[Co(en)_2Cl_2]^+$ complex ion has an octahedral geometry.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) (A) is correct but (R) is not correct

(B) (A) is not correct but (R) is correct

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Q145. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

Match List I with List II

	LIST I (Complex ion)		LIST II (Spin only magnetic moment in B.M.)
A.	$[Cr(NH_3)_6]^{3+}$	I.	4.90
B.	$[NiCl_4]^{2-}$	II.	3.87
C.	$[CoF_6]^{3-}$	III.	0.0
D.	$[Ni(CN)_4]^{2-}$	IV.	2.83

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(B) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(C) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

(D) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Q146. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Number of Complexes with even number of electrons in t_{2g} orbitals is -

$[Fe(H_2O)_6]^{2+}, [Co(H_2O)_6]^{2+}, [Co(H_2O)_6]^{3+}, [Cu(H_2O)_6]^{2+}, [Cr(H_2O)_6]^{2+}$

(A) 3 **(B)** 2 **(C)** 1 **(D)** 5

Q147. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

An octahedral complex with the formula $CoCl_3 \cdot nNH_3$ upon reaction with excess of $AgNO_3$ solution gives 2 moles of $AgCl$. Consider the oxidation state of Co in the complex is 'x'. The value of "$x + n$" is _____.

(A) 6 **(B)** 5 **(C)** 3 **(D)** 8

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Q148. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Given below are two statements:

Statement I: $N(CH_3)_3$ and $P(CH_3)_3$ can act as ligands to form transition metal complexes.

Statement II: As N and P are from same group, the nature of bonding of $N(CH_3)_3$ and $P(CH_3)_3$ is always same with transition metals.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are incorrect.

(B) Both Statement I and Statement II are correct.

(C) Statement I is incorrect but Statement II is correct.

(D) Statement I is correct but Statement II is incorrect.

Q149. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Match List I with List II

	LIST I (Compound)		LIST II (Colour)
A.	$Fe_4[Fe(CN)_6]_3 \cdot xH_2O$	I.	Violet
B.	$[Fe(CN)_5NOS]^{4-}$	II.	Blood Red
C.	$[Fe(SCN)]^{2+}$	III.	Prussian Blue
D.	$(NH_4)_3PO_4 \cdot 12MoO_3$	IV.	Yellow

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-I, B-II, C-III, D-IV

(C) A-II, B-III, C-IV, D-I

(D) A-III, B-I, C-II, D-IV

Q150. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

Given below are two statements :

Statement I : PF_5 and BrF_5 both exhibit $sp^3\,d$ hybridisation.

Statement II : Both SF_6 and $[Co(NH_3)_6]^{3+}$ exhibit $sp^3\,d^2$ hybridisation.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

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Q151. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

Match List I with List II.

	LIST I Tetrahedral Complex		LIST II Electronic configuration
A.	$TiCl_4$	I.	e^2, t_2^0
B.	$[FeO_4]^{2-}$	II.	e^4, t_2^3
C.	$[FeCl_4]^-$	III.	e^0, t_2^0
D.	$[CoCl_4]^{2-}$	IV.	e^2, t_2^3

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(B) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(C) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q152. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

The correct IUPAC name of $[PtBr_2(PMe_3)_2]$ is :

(A) bis(trimethylphosphine)dibromoplatinum(II)

(B) dibromodi(trimethylphosphine)platinum(II)

(C) dibromobis(trimethylphosphine)platinum(II)

(D) bis[bromo(trimethylphosphine)]platinum(II)

Q153. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

Consider the following complexes

(A) $[CoCl(NH_3)_5]^{2+}$, (B) $[Co(CN)_6]^{3-}$, (C) $[Co(NH_3)_5(H_2O)]^{3+}$, (D) $[Cu(H_2O)_4]^{2+}$

The correct order of A, B, C and D in terms of wavenumber of light absorbed is :

(A) D < A < C < B

(B) B < C < A < D

(C) C < D < A < B

(D) A < C < B < D

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Q154. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

Match List I with List II

	LIST I (Hybridization)		LIST II (Orientation in Shape)
A.	sp^3	I.	Trigonal bipyramidal
B.	dsp^2	II.	Octahedral
C.	sp^3d	III.	Tetrahedral
D.	sp^3d^2	IV.	Square planar

Choose the correct answer from the options given below:

(A) A-III, B-I, C-IV, D-II
(B) A-III, B-IV, C-I, D-II
(C) A-IV, B-III, C-I, D-II
(D) A-II, B-I, C-IV, D-III

Q155. JEE Main 2024 (Online) 5th April Evening Shift

Open solution

The number of complexes from the following with no electrons in the t_2 orbital is _____.

$TiCl_4, [MnO_4]^-, [FeO_4]^{2-}, [FeCl_4]^-, [CoCl_4]^{2-}$

(A) 4
(B) 2
(C) 3
(D) 1

Q156. JEE Main 2024 (Online) 5th April Evening Shift

Open solution

The metal atom present in the complex MABXL (where A, B, X and L are unidentate ligands and M is metal) involves sp^3 hybridization. The number of geometrical isomers exhibited by the complex is :

(A) 0
(B) 2
(C) 3
(D) 4

Q157. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

The correct order of ligands arranged in increasing field strength.

(A) $F^- < Br^- < I^- < NH_3$
(B) $H_2O < {}^-OH < CN^- < NH_3$
(C) $Br^- < F^- < H_2O < NH_3$
(D) $Cl^- < {}^-OH < Br^- < CN^-$

Q158. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Which one of the following complexes will exhibit the least paramagnetic behaviour ? [Atomic number, $Cr = 24, Mn = 25, Fe = 26, Co = 27$]

(A) $[Fe(H_2O)_6]^{2+}$
(B) $[Mn(H_2O)_6]^{2+}$
(C) $[Co(H_2O)_6]^{2+}$
(D) $[Cr(H_2O)_6]^{2+}$

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Q159. JEE Main 2024 (Online) 4th April Evening Shift

If an iron (III) complex with the formula $[Fe(NH_3)_x(CN)_y]^-$ has no electron in its e_g orbital, then the value of $x + y$ is

(A) 6 (B) 4 (C) 3 (D) 5

Open solution

Q160. JEE Main 2024 (Online) 4th April Evening Shift

The number of unpaired d-electrons in $[Co(H_2O)_6]^{3+}$ is _____.

(A) 0 (B) 2 (C) 1 (D) 4

Open solution

Q161. JEE Main 2024 (Online) 4th April Evening Shift

A first row transition metal in its +2 oxidation state has a spin-only magnetic moment value of $3.86\ BM$. The atomic number of the metal is

(A) 22 (B) 23 (C) 26 (D) 25

Open solution

Q162. JEE Main 2024 (Online) 4th April Morning Shift

Number of complexes from the following with even number of unpaired "d" electrons is _____
$[V(H_2O)_6]^{3+}, [Cr(H_2O)_6]^{2+}, [Fe(H_2O)_6]^{3+}, [Ni(H_2O)_6]^{3+}, [Cu(H_2O)_6]^{2+}$ [Given atomic numbers: $V = 23, Cr = 24, Fe = 26, Ni = 28, Cu = 29$]

(A) 1 (B) 5 (C) 2 (D) 4

Open solution

Q163. JEE Main 2024 (Online) 4th April Morning Shift

The correct sequence of ligands in the order of decreasing field strength is :

(A) $NCS^- > EDTA^{4-} > CN^- > CO$ (B) $CO > H_2O > F^- > S^{2-}$

(C) $S^{2-} > {}^-OH > EDTA^{4-} > CO$ (D) ${}^-OH > F^- > NH_3 > CN^-$

Open solution

Q164. JEE Main 2024 (Online) 1st February Evening Shift

$[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ are respectively known as :

(A) Inner orbital Complex, Spin paired Complex

(B) Spin paired Complex, Spin free Complex

(C) Spin free Complex, Spin paired Complex

(D) Outer orbital Complex, Inner orbital Complex

Open solution

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Q165. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Given below are two statements :

Statement (I) : Dimethyl glyoxime forms a six-membered covalent chelate when treated with $NiCl_2$ solution in presence of NH_4OH.

Statement (II) : Prussian blue precipitate contains iron both in $(+2)$ and $(+3)$ oxidation states.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Q166. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Which of the following compounds show colour due to d-d transition?

(A) $K_2Cr_2O_7$

(B) $CuSO_4 \cdot 5H_2O$

(C) $KMnO_4$

(D) K_2CrO_4

Q167. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

Which of the following complex is homoleptic?

(A) $[Ni(NH_3)_2Cl_2]$

(B) $[Co(NH_3)_4Cl_2]^+$

(C) $[Fe(NH_3)_4Cl_2]^+$

(D) $[Ni(CN)_4]^{2-}$

Q168. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

Given below are two statements :

Statement (I) : A solution of $[Ni(H_2O)_6]^{2+}$ is green in colour.

Statement (II) : A solution of $[Ni(CN)_4]^{2-}$ is colourless.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

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Q169. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Select the option with correct property -

(A) $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ both Paramagnetic

(B) $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ both Diamagnetic

(C) $[NiCl_4]^{2-}$ Diamagnetic, $[Ni(CO)_4]$ Paramagnetic

(D) $[Ni(CO)_4]$ Diamagnetic, $[NiCl_4]^{2-}$ Paramagnetic

Q170. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Match List I with List II

	List - I (Complex ion)		List - II (Electronic Configuration)
(A)	$[Cr(H_2O)_6]^{3+}$	(I)	$t_{2g}^2 e_g^0$
(B)	$[Fe(H_2O)_6]^{3+}$	(II)	$t_{2g}^3 e_g^0$
(C)	$[Ni(H_2O)_6]^{2+}$	(III)	$t_{2g}^3 e_g^2$
(D)	$[V(H_2O)_6]^{3+}$	(IV)	$t_{2g}^6 e_g^2$

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-III, B-II, C-IV, D-I

(C) A-II, B-III, C-IV, D-I

(D) A-IV, B-III, C-I, D-II

Q171. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

The correct statements from following are:

A. The strength of anionic ligands can be explained by crystal field theory.

B. Valence bond theory does not give a quantitative interpretation of kinetic stability of coordination compounds.

C. The hybridization involved in formation of $[Ni(CN)_4]^{2-}$ complex is dsp^2.

D. The number of possible isomer(s) of cis- $[PtCl_2(en)_2]^{2+}$ is one

Choose the correct answer from the options given below:

(A) B, C only

(B) B, D only

(C) A, C only

(D) A, D only

Q172. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

The coordination geometry around the manganese in decacarbonyldimanganese (0) is

(A) Trigonal bipyramidal

(B) Square pyramidal

(C) Square planar

(D) Octahedral

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Q173. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Choose the correct statements from the following :

(A) Ethane-1, 2-diamine is a chelating ligand.

(B) Metallic aluminium is produced by electrolysis of aluminium oxide in presence of cryolite.

(C) Cyanide ion is used as ligand for leaching of silver.

(D) Phosphine acts as a ligand in Wilkinson catalyst.

(E) The stability constants of Ca^{2+} and Mg^{2+} are similar with EDTA complexes.

Choose the correct answer from the options given below :

(A) (B), (C), (E) only

(B) (A), (D), (E) only

(C) (C), (D), (E) only

(D) (A), (B), (C) only

Q174. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

A reagent which gives brilliant red precipitate with Nickel ions in basic medium is

(A) dimethyl glyoxime

(B) sodium nitroprusside

(C) meta-dinitrobenzene

(D) neutral $FeCl_3$

Q175. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Match List I with List II

	List - I (Substances)		List - II (Element Present)
(A)	Ziegler catalyst	(I)	Rhodium
(B)	Blood Pigment	(II)	Cobalt
(C)	Wilkinson catalyst	(III)	Iron
(D)	Vitamin B_{12}	(IV)	Titanium

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II

(B) A-II, B-IV, C-I, D-III

(C) A-III, B-II, C-IV, D-I

(D) A-II, B-III, C-IV, D-I

Q176. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

In which one of the following metal carbonyls, CO forms a bridge between metal atoms?

(A) $[Os_3(CO)_{12}]$

(B) $[Ru_3(CO)_{12}]$

(C) $[Mn_2(CO)_{10}]$

(D) $[Co_2(CO)_8]$

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Q177. JEE Main 2024 (Online) 27th January Evening Shift

Identify the incorrect pair from the following :

(A) Haber process - Iron

(B) Polythene preparation - $TiCl_4, Al(CH_3)_3$

(C) Photography - AgBr

(D) Wacker process - $PtCl_2$

Open solution

Q178. JEE Main 2024 (Online) 27th January Evening Shift

Identify from the following species in which d^2sp^3 hybridization is shown by central atom :

(A) $[Co(NH_3)_6]^{3+}$

(B) SF_6

(C) $[Pt\,(Cl_4)]^{2-}$

(D) BrF_5

Open solution

Q179. JEE Main 2024 (Online) 27th January Morning Shift

Consider the following complex ions

$P = [FeF_6]^{3-}$

$Q = [V(H_2O)_6]^{2+}$

$R = [Fe(H_2O)_6]^{2+}$

The correct order of the complex ions, according to their spin only magnetic moment values (in B.M.) is :

(A) R < Q < P

(B) R < P < Q

(C) Q < R < P

(D) Q < P < R

Open solution

Q180. JEE Main 2024 (Online) 27th January Morning Shift

Yellow compound of lead chromate gets dissolved on treatment with hot $NaOH$ solution. The product of lead formed is a :

(A) Tetraanionic complex with coordination number six

(B) Neutral complex with coordination number four

(C) Dianionic complex with coordination number six

(D) Dianionic complex with coordination number four

Open solution

Q181. JEE Main 2023 (Online) 15th April Morning Shift

The complex with highest magnitude of crystal field splitting energy (Δ_0) is :

(A) $[Fe(OH_2)_6]^{3+}$

(B) $[Mn(OH_2)_6]^{3+}$

(C) $[Cr(OH_2)_6]^{3+}$

(D) $[Ti(OH_2)_6]^{3+}$

Open solution

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Q182. JEE Main 2023 (Online) 13th April Evening Shift

The total number of stereoisomers for the complex $[Cr(ox)_2ClBr]^{3-}$ (where ox = oxalate) is :

(A) 1 (B) 3 (C) 2 (D) 4

Open solution

Q183. JEE Main 2023 (Online) 13th April Evening Shift

Which of the following complexes will exhibit maximum attraction to an applied magnetic field?

(A) $[Ni(H_2O)_6]^{2+}$ (B) $[Co(H_2O)_6]^{2+}$

(C) $[Co(en)_3]^{3+}$ (D) $[Zn(H_2O)_6]^{2+}$

Open solution

Q184. JEE Main 2023 (Online) 13th April Morning Shift

The mismatched combinations are

A. Chlorophyll - Co

B. Water hardness - EDTA

C. Photography $-[Ag(CN)_2]^-$

D. Wilkinson catalyst $-[(Ph_3P)_3RhCl]$

E. Chelating ligand - D-Penicillamine

Choose the correct answer from the options given below :

(A) A and E Only (B) D and E Only (C) A and C Only (D) A, C, and E Only

Open solution

Q185. JEE Main 2023 (Online) 12th April Morning Shift

Match List I with List II

	LIST I Complex		**LIST II CFSE (Δ_0)**
A.	$[Cu(NH_3)_6]^{2+}$	I.	-0.6
B.	$[Ti(H_2O)_6]^{3+}$	II.	-2.0
C.	$[Fe(CN)_6]^{3-}$	III.	-1.2
D.	$[NiF_6]^{4-}$	IV.	-0.4

Choose the correct answer from the options given below:

(A) A-I, B-IV, C-II, D-III (B) A-III, B-IV, C-I, D-II

(C) A-II, B-III, C-I, D-IV (D) A-I, B-II, C-IV, D-III

Open solution

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Q186. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : $[CoCl(NH_3)_5]^{2+}$ absorbs at lower wavelength of light with respect to $[Co(NH_3)_5\,(H_2O)]^{3+}$

Reason R : It is because the wavelength of the light absorbed depends on the oxidation state of the metal ion.

In the light of the above statements, choose the correct answer from the options given below:

(A) A is false but R is true

(B) Both A and R are true but R is NOT the correct explanation of A

(C) Both A and R are true and R is the correct explanation of A

(D) A is true but R is false

Q187. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

If Ni^{2+} is replaced by Pt^{2+} in the complex $[NiCl_2Br_2]^{2-}$, which of the following properties are expected to get changed ?

A. Geometry

B. Geometrical isomerism

C. Optical isomerism

D. Magnetic properties

(A) A and D **(B)** A, B and D **(C)** A, B and C **(D)** B and C

Q188. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

Match List I with List II

	LIST I Complex		LIST II Colour
A.	$Mg(NH_4)PO_4$	I.	brown
B.	$K_3[Co(NO_2)_6]$	II.	white
C.	$MnO(OH)_2$	III.	yellow
D.	$Fe_4[Fe(CN)_6]_3$	IV.	blue

Choose the correct answer from the options given below :

(A) A-II, B-IV, C-I, D-III **(B)** A-II, B-III, C-I, D-IV

(C) A-III, B-IV, C-II, D-I **(D)** A-II, B-III, C-IV, D-I

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Q189. JEE Main 2023 (Online) 11th April Evening Shift

The magnetic moment is measured in Bohr Magneton (BM).

Spin only magnetic moment of Fe in $[Fe(H_2O)_6]^{3+}$ and $[Fe(CN)_6]^{3-}$ complexes respectively is :

(A) 3.87 B.M. and 1.732 B.M.

(B) 5.92 B.M. and 1.732 B.M.

(C) 6.92 B.M. in both

(D) 4.89 B.M. and 6.92 B.M.

Open solution

Q190. JEE Main 2023 (Online) 11th April Morning Shift

The complex that dissolves in water is :

(A) $Fe_4[Fe(CN)_6]_3$

(B) $(NH_4)_3\,[As(Mo_3O_{10})_4]$

(C) $K_3\,[Co(NO_2)_6]$

(D) $[Fe_3(OH)_2(OAc)_6]Cl$

Open solution

Q191. JEE Main 2023 (Online) 11th April Morning Shift

The set which does not have ambidentate ligand(s) is :

(A) ${C_2O_4}^{2-}, {NO_2}^{-}, NCS^{-}$

(B) ${C_2O_4}^{2-}$, ethylene diamine, H_2O

(C) NO_2^{-}, ${C_2O_4}^{2-}$, EDTA $^{4-}$

(D) $EDTA^{4-}, NCS^{-}, {C_2O_4}^{2-}$

Open solution

Q192. JEE Main 2023 (Online) 11th April Morning Shift

Which of the following complex has a possibility to exist as meridional isomer?

(A) $[Pt(NH_3)_2Cl_2]$

(B) $[Co(en)_2Cl_2]$

(C) $[Co(en)_3]$

(D) $[Co(NH_3)_3(NO_2)_3]$

Open solution

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Q193. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

The correct order of the number of unpaired electrons in the given complexes is

A. $[\mathrm{Fe(CN)_6}]^{3-}$

B. $[\mathrm{FeF_6}]^{3-}$

C. $[\mathrm{CoF_6}]^{3-}$

D. $[\mathrm{Cr\ (oxalate)_3}]^{3-}$

E. $[\mathrm{Ni(CO)_4}]$

Choose the correct answer from the options given below:

(A) E < A < B < D < C

(B) A < E < D < C < B

(C) A < E < C < B < D

(D) E < A < D < C < B

Q194. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

Match List I with List II

	List - I Complex		List - II Crystal Field splitting energy (Δ_0)
A.	$[Ti(H_2O)_6]^{2+}$	I.	-1.2
B.	$[V(H_2O)_6]^{2+}$	II.	-0.6
C.	$[Mn(H_2O)_6]^{3+}$	III.	0
D.	$[Fe(H_2O)_6]^{3+}$	IV.	-0.8

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-II, B-IV, C-III, D-I

(C) A-II, B-IV, C-I, D-III

(D) A-IV, B-I, C-III, D-II

Q195. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The octahedral diamagnetic low spin complex among the following is :

(A) $[Co(NH_3)_6]^{3+}$ **(B)** $[NiCl_4]^{2-}$ **(C)** $[CoCl_6]^{3-}$ **(D)** $[CoF_6]^{3-}$

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Q196. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

Match List I with List II

	LIST I Coordination Complex		LIST II Number of unpaired electrons
A.	$[Cr(CN)_6]^{3-}$	I.	0
B.	$[Fe(H_2O)_6]^{2+}$	II.	3
C.	$[Co(NH_3)_6]^{3+}$	III.	2
D.	$[Ni(NH_3)_6]^{2+}$	IV.	4

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-II, D-I

(B) A-II, B-I, C-IV, D-III

(C) A-II, B-IV, C-I, D-III

(D) A-III, B-IV, C-I, D-II

Q197. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

Which of the following complex is octahedral, diamagnetic and the most stable?

(A) $Na_3[CoCl_6]$

(B) $[Co(H_2O)_6]Cl_2$

(C) $K_3[Co(CN)_6]$

(D) $[Ni(NH_3)_6]Cl_2$

Q198. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

The correct order of spin only magnetic moments for the following complex ions is

(A) $[Fe(CN)_6]^{3-} < [CoF_6]^{3-} < [MnBr_4]^{2-} < [Mn(CN)_6]^{3-}$

(B) $[MnBr_4]^{2-} < [CoF_6]^{3-} < [Fe(CN)_6]^{3-} < [Mn(CN)_6]^{3-}$

(C) $[Fe(CN)_6]^{3-} < [Mn(CN)_6]^{3-} < [CoF_6]^{3-} < [MnBr_4]^{2-}$

(D) $[CoF_6]^{3-} < [MnBr_4]^{2-} < [Fe(CN)_6]^{3-} < [Mn(CN)_6]^{3-}$

Q199. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

The IUPAC name of $K_3[Co(C_2O_4)_3]$ is:-

(A) Potassium tris(oxalato)cobalt(III)

(B) Potassium tris(oxalato)cobaltate(III)

(C) Potassium trioxalatocobaltate(III)

(D) Potassium trioxalatocobalt(III)

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Q200. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

Given below are two statements, one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$.

Assertion A: The spin only magnetic moment value for $[Fe(CN)_6]^{3-}$ is $1.74BM$, whereas for $[Fe(H_2O)_6]^{3+}$ is $5.92BM$.

Reason $\mathbf{R}$: In both complexes, Fe is present in +3 oxidation state.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both A and R are true and R is the correct explanation of A

(B) A is true but R is false

(C) A is false but R is true

(D) Both A and R are true but R is NOT the correct explanation of A

Q201. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

The complex cation which has two isomers is :

(A) $[Co(NH_3)_5Cl]^+$

(B) $[Co(H_2O)_6]^{3+}$

(C) $[Co(NH_3)_5NO_2]^{2+}$

(D) $[Co(NH_3)_5Cl]^{2+}$

Q202. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

Which of the following complexes will show largest splitting of d-orbitals?

(A) $[Fe(C_2O_4)_3]^{3-}$

(B) $[Fe(CN)_6]^{3-}$

(C) $[Fe(NH_3)_6]^{3+}$

(D) $[FeF_6]^{3-}$

Q203. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

Which of the following are the example of double salt?

A. $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$

B. $CuSO_4 \cdot 4NH_3 \cdot H_2O$

C. $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

D. $Fe(CN)_2 \cdot 4KCN$

Choose the correct answer :

(A) A, B and D only

(B) B and D only

(C) A and B only

(D) A and C only

Q204. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

A solution of $FeCl_3$ when treated with $K_4[Fe(CN)_6]$ gives a Prussian blue precipitate due to the formation of :

(A) $Fe[Fe(CN)_6]$

(B) $Fe_4[Fe(CN)_6]_3$

(C) $Fe_3[Fe(CN)_6]_2$

(D) $K[Fe_2(CN)_6]$

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Q205. JEE Main 2023 (Online) 31st January Morning Shift

Cobalt chloride when dissolved in water forms pink colored complex $\underline{X}$ which has octahedral geometry. This solution on treating with conc HCl forms deep blue complex, $\underline{Y}$ which has a $\underline{Z}$ geometry. X, Y and Z, respectively, are

Open solution

(A) $X = [Co(H_2O)_6]^{2+}, Y = [CoCl_4]^{2-}, Z =$ Tetrahedral

(B) $X = [Co(H_2O)_6]^{2+}, Y = [CoCl_6]^{3-}, Z =$ Octahedral

(C) $X = [Co(H_2O)_4Cl_2]^{+}, Y = [CoCl_4]^{2-}, Z =$ Tetrahedral

(D) $X = [Co(H_2O)_6]^{3+}, Y = [CoCl_6]^{3-}, Z =$ Octahedral

Q206. JEE Main 2023 (Online) 30th January Evening Shift

Match List **I** with List **II**:

Open solution

List I (Complexes)		List II (Hybridisation)	
A.	$[Ni(CO)_4]$	I.	sp^3
B.	$[Cu(NH_3)_4]^{2+}$	II.	dsp^2
C.	$[Fe(NH_3)_6]^{2+}$	III.	sp^3d^2
D.	$[Fe(H_2O)_6]^{2+}$	IV.	d^2sp^3

(A) A-I, B-II, C-III, D-IV

(B) A-II, B-I, C-III, D-IV

(C) A-I, B-II, C-IV, D-III

(D) A-II, B-I, C-IV, D-III

Q207. JEE Main 2023 (Online) 30th January Evening Shift

The $Cl - Co - Cl$ bond angle values in a fac- $[Co(NH_3)_3Cl_3]$ complex is/are :

Open solution

(A) 90° & 120° (B) 90° (C) 180° (D) 90° & 180°

Q208. JEE Main 2023 (Online) 30th January Evening Shift

$1\,L, 0.02M$ solution of $[Co(NH_3)_5SO_4]$ Br is mixed with $1\,L, 0.02M$ solution of $[Co(NH_3)_5Br]SO_4$. The resulting solution is divided into two equal parts (X) and treated with excess of $AgNO_3$ solution and $BaCl_2$ solution respectively as shown below:

Open solution

$1\,L$ Solution $(X) + AgNO_3$ solution (excess) $\longrightarrow Y$

$1\,L$ Solution $(X) + BaCl_2$ solution (excess) $\longrightarrow Z$

The number of moles of Y and Z respectively are

(A) $0.01, 0.01$ (B) $0.01, 0.02$ (C) $0.02, 0.01$ (D) $0.02, 0.02$

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Q209. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

To inhibit the growth of tumours, identify the compounds used from the following :

A. EDTA

B. Coordination Compounds of Pt

C. D - Penicillamine

D. Cis - Platin

Choose the correct answer from the option given below :

(A) A and B Only **(B)** C and D Only **(C)** B and D Only **(D)** A and C Only

Q210. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Which of the following is correct order of ligand field strength?

(A) $S^{2-} < C_2O_4^{2-} < NH_3 < en < CO$

(B) $CO < en < NH_3 < C_2O_4^{2-} < S^{2-}$

(C) $S^{2-} < NH_3 < en < CO < C_2O_4^{2-}$

(D) $NH_3 < en < CO < S^{2-} < C_2O_4^{2-}$

Q211. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Correct order of spin only magnetic moment of the following complex ions is :

(Given At.no. Fe : 26, Co : 27)

(A) $[CoF_6]^{3-} > [FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$

(B) $[FeF_6]^{3-} > [Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-}$

(C) $[FeF_6]^{3-} > [CoF_6]^{3-} > [Co(C_2O_4)_3]^{3-}$

(D) $[Co(C_2O_4)_3]^{3-} > [CoF_6]^{3-} > [FeF_6]^{3-}$

Q212. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Chiral complex from the following is :

Here en = ethylene diamine

(A) $cis - [PtCl_2(en)_2]^{2+}$

(B) $trans - [Co(NH_3)_4Cl_2]^{+}$

(C) $trans - [PtCl_2(en)_2]^{2+}$

(D) $cis - [PtCl_2(NH_3)_2]$

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Q213. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Match List I with List II

	List I Coordination entity		List II Wavelength of light absorbed in nm
A.	$[CoCl(NH_3)_5]^{2+}$	I.	310
B.	$[Co(NH_3)_6]^{3+}$	II.	475
C.	$[Co(CN)_6]^{3-}$	III.	535
D.	$[Cu(H_2O)_4]^{2+}$	IV.	600

Choose the correct answer from the options given below :

(A) A-IV, B-I, C-III, D-II

(B) A-III, B-II, C-I, D-IV

(C) A-III, B-I, C-II, D-IV

(D) A-II, B-III, C-IV, D-I

Q214. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

The hybridization and magnetic behaviour of cobalt ion in $[Co(NH_3)_6]^{3+}$ complex, respectively is :

(A) sp^3d^2 and paramagnetic

(B) sp^3d^2 and diamagnetic

(C) d^2sp^3 and diamagnetic

(D) d^2sp^3 and paramagnetic

Q215. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

Which of the following cannot be explained by crystal field theory?

(A) The order of spectrochemical series

(B) Stability of metal complexes

(C) Magnetic properties of transition metal complexes

(D) Colour of metal complexes

Q216. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

The primary and secondary valencies of cobalt respectively in $[Co(NH_3)_5Cl]Cl_2$ are :

(A) 2 and 8

(B) 2 and 6

(C) 3 and 5

(D) 3 and 6

Q217. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

In low and high spin octahedral Co^{3+} complexes the number of electrons in t_{2g} level respectively will be . (given At. no. of Co, 27)

(A) 6 and 3

(B) 6 and 4

(C) 3 and 4

(D) 4 and 6

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Q218. JEE Main 2022 (Online) 30th July Evening Shift

In the ring test of NO_3^- ion what is the oxidation number of iron in the complex (brown ring) $[Fe(H_2O)_5NO]SO_4$?

(A) $+2$ (B) $+1$ (C) $+3$ (D) $+5$

Open solution

Q219. JEE Main 2022 (Online) 29th July Evening Shift

Octahedral complexes of copper(II) undergo structural distortion (Jahn-Teller). Which one of the given copper (II) complexes will show the maximum structural distortion? (en - ethylenediamine; $H_2N-CH_2-CH_2-NH_2$)

(A) $[Cu(H_2O)_6]SO_4$

(B) $[Cu(en)(H_2O)_4]SO_4$

(C) cis-$[Cu(en)_2Cl_2]$

(D) trans-$[Cu(en)_2Cl_2]$

Open solution

Q220. JEE Main 2022 (Online) 28th July Evening Shift

Match List I with List II

	List - I (Complex)		List - II (Hybridization)
(A)	$Ni(CO)_4$	(I)	sp^3
(B)	$[Ni(CN)_4]^{2-}$	(II)	sp^3d^2
(C)	$[Co(CN)_6]^{3-}$	(III)	d^2sp^3
(D)	$[CoF_6]^{3-}$	(IV)	dsp^2

Choose the correct answer from the options given below :

(A) A-IV, B-I, C-III, D-II (B) A-I, B-IV, C-III, D-II

(C) A-I, B-IV, C-II, D-III (D) A-IV, B-I, C-II, D-III

Open solution

Q221. JEE Main 2022 (Online) 27th July Evening Shift

Low oxidation state of metals in their complexes are common when ligands :

(A) have good π-accepting character (B) have good σ-donor character

(C) are having good π-donating ability (D) are having poor σ-donating ability

Open solution

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Q222. JEE Main 2022 (Online) 27th July Evening Shift

Fe^{3+} cation gives a prussian blue precipitate on addition of potassium ferrocyanide solution due to the formation of :

(A) $[Fe(H_2O)_6]_2\,[Fe(CN)_6]$

(B) $Fe_2[Fe(CN)_6]_2$

(C) $Fe_3[Fe(OH)_2(CN)_4]_2$

(D) $Fe_4[Fe(CN)_6]_3$

Open solution

Q223. JEE Main 2022 (Online) 26th July Evening Shift

The metal complex that is diamagnetic is (Atomic number: $Fe, 26; Cu, 29$)

(A) $K_3\,[Cu(CN)_4]$

(B) $K_2\,[Cu(CN)_4]$

(C) $K_3\,[Fe(CN)_4]$

(D) $K_4\,[FeCl_6]$

Open solution

Q224. JEE Main 2022 (Online) 25th July Evening Shift

The correct order of energy of absorption for the following metal complexes is :

A : $[Ni(en)_3]^{2+}$, B : $[Ni(NH_3)_6]^{2+}$, C : $[Ni(H_2O)_6]^{2+}$

(A) C < B < A

(B) B < C < A

(C) C < A < B

(D) A < C < B

Open solution

Q225. JEE Main 2022 (Online) 30th June Morning Shift

Correct formula of the compound which gives a white precipitate with $BaCl_2$ solution, but not with $AgNO_3$ solution, is :

(A) $[Co(NH_3)_5Br]SO_4$

(B) $[Co(NH_3)_5SO_4]Br$

(C) $[Pt(NH_3)_4Cl_2]Br_2$

(D) $[Pt(NH_3)_4Br_2]Cl_2$

Open solution

Q226. JEE Main 2022 (Online) 29th June Evening Shift

Given below are two statements.

- Statement I : In $CuSO_4 \cdot 5H_2O$, Cu-O bonds are present.
- Statement II : In $CuSO_4 \cdot 5H_2O$, ligands coordinating with Cu(II) ion are O-and S-based ligands.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Open solution

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Q227. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Given below are two statements :

Statement I : $[Ni(CN)_4]^{2-}$ is square planar and diamagnetic complex, with dsp^2 hybridization for Ni but $[Ni(CO)_4]$ is tetrahedral, paramagnetic and with sp^3-hybridization for Ni.

Statement II : $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ both have same d-electron configuration have same geometry and are paramagnetic.

In light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is correct but Statement II is false.

(D) Statement I is false but Statement II is correct.

Q228. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

Arrange the following coordination compounds in the increasing order of magnetic moments.
(Atomic numbers : Mn = 25; Fe = 26)

A. $[FeF_6]^{3-}$

B. $[Fe(CN)_6]^{3-}$

C. $[MnCl_6]^{3-}$ (high spin)

D. $[Mn(CN)_6]^{3-}$

Choose the correct answer from the options given below :

(A) A < B < D < C **(B)** B < D < C < A **(C)** A < C < D < B **(D)** B < D < A < C

Q229. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

Which of the following will have maximum stabilization due to crystal field?

(A) $[Ti(H_2O)_6]^{3+}$ **(B)** $[Co(H_2O)_6]^{2+}$ **(C)** $[Co(CN)_6]^{3-}$ **(D)** $[Cu(NH_3)_4]^{2+}$

Q230. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

Which statement is not true with respect to nitrate ion test?

(A) A dark brown ring is formed at the junction of two solutions.

(B) Ring is formed due to nitroferrous sulphate complex.

(C) The brown complex is $[Fe(H_2O)_5(NO)]SO_4$.

(D) Heating the nitrate salt with conc. H_2SO_4, light brown fumes are evolved.

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Q231. JEE Main 2022 (Online) 25th June Morning Shift

White precipitate of AgCl dissolves in aqueous ammonia solution due to formation of :

(A) $[Ag(NH_3)_4]Cl_2$ (B) $[Ag(Cl)_2(NH_3)_2]$ (C) $[Ag(NH_3)_2]Cl$ (D) $[Ag(NH_3)Cl]Cl$

Open solution

Q232. JEE Main 2022 (Online) 24th June Evening Shift

Transition metal complex with highest value of crystal field splitting (Δ_0) will be :

(A) $[Cr(H_2O)_6]^{3+}$ (B) $[Mo(H_2O)_6]^{3+}$ (C) $[Fe(H_2O)_6]^{3+}$ (D) $[Os(H_2O)_6]^{3+}$

Open solution

Q233. JEE Main 2022 (Online) 24th June Morning Shift

Match List - I with List - II :

	List - I		List -II
(A)	$[PtCl_4]^{2-}$	(I)	sp^3d
(B)	BrF_5	(II)	d^2sp^3
(C)	PCl_5	(III)	dsp^2
(D)	$[Co(NH_3)_6]^{3+}$	(IV)	sp^3d^2

Choose the most appropriate answer from the options given below :

(A) (A)-(II), (B)-(IV), (C)-(I), (D)-(III) (B) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) (D) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Open solution

Q234. JEE Main 2021 (Online) 1st September Evening Shift

The Crystal Field Stabilization Energy (CFSE) and magnetic moment (spin-only) of an octahedral aqua complex of a metal ion (M^{z+}) are $-0.8\ \Delta_0$ and 3.87 BM, respectively. Identify (M^{z+}) :

(A) V^{3+} (B) Cr^{3+} (C) Mn^{4+} (D) Co^{2+}

Open solution

Q235. JEE Main 2021 (Online) 1st September Evening Shift

The potassium ferrocyanide solution gives a Prussian blue colour, when added to :

(A) $CoCl_3$ (B) $FeCl_2$ (C) $CoCl_2$ (D) $FeCl_3$

Open solution

Q236. JEE Main 2021 (Online) 31st August Evening Shift

Spin only magnetic moment in BM of $[Fe(CO)_4(C_2O_4)]^+$ is :

(A) 5.92 (B) 0 (C) 1 (D) 1.73

Open solution

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Q237. JEE Main 2021 (Online) 31st August Morning Shift

The denticity of an organic ligand, biuret is :

(A) 2 (B) 4 (C) 3 (D) 6

Open solution

Q238. JEE Main 2021 (Online) 26th August Evening Shift

Indicate the complex/complex ion which did not show any geometrical isomerism :

(A) $[CoCl_2(en)_2]$ (B) $[Co(CN)_5(NC)]^{3-}$ (C) $[Co(NH_3)_3(NO_2)_3]$ (D) $[Co(NH_3)_4Cl_2]^+$

Open solution

Q239. JEE Main 2021 (Online) 26th August Evening Shift

Arrange the following Cobalt complexes in the order of increasing Crystal Field Stabilization Energy (CFSE) value.

Complexes : $\underset{A}{[CoF_6]^{3-}}, \underset{B}{[Co(H_2O)_6]^{2+}}, \underset{C}{[Co(NH_3)_6]^{3+}} and \underset{D}{[Co(en)_3]^{3+}}$

Choose the correct option :

(A) A < B < C < D (B) B < A < C < D (C) B < C < D < A (D) C < D < B < A

Open solution

Q240. JEE Main 2021 (Online) 27th July Evening Shift

Given below are two statements :

Statement I : $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$ are d^2sp^3 hybridised.

Statement II : $[MnCl_6]^{3-}$ and $[FeF_6]^{3-}$ are paramagnetic and have 4 and 5 unpaired electrons, respectively.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is correct but statement II is false

(B) Both statement I and Statement II are false

(C) Statement I is incorrect but statement II is true

(D) Both statement I and statement II are true

Open solution

Q241. JEE Main 2021 (Online) 27th July Morning Shift

The type of hybridisation and magnetic property of the complex $[MnCl_6]^{3-}$, respectively, are :

(A) sp^3d^2 and diamagnetic (B) d^2sp^3 and diamagnetic

(C) d^2sp^3 and paramagnetic (D) sp^3d^2 and paramagnetic

Open solution

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Q242. JEE Main 2021 (Online) 27th July Morning Shift

The number of geometrical isomers found in the metal complexes $[PtCl_2(NH_3)_2]$, $[Ni(CO)_4]$, $[Ru(H_2O)_3Cl_3]$ and $[CoCl_2(NH_3)_4]^+$ respectively, are :

(A) 1, 1, 1, 1 (B) 2, 1, 2, 2 (C) 2, 0, 2, 2 (D) 2, 1, 2, 1

Open solution

Q243. JEE Main 2021 (Online) 25th July Evening Shift

Which one of the following metal complexes is most stable?

(A) $[Co(en)(NH_3)_4]Cl_2$ (B) $[Co(en)_3]Cl_2$

(C) $[Co(en)_2(NH_3)_2]Cl_2$ (D) $[Co(NH_3)_6]Cl_2$

Open solution

Q244. JEE Main 2021 (Online) 25th July Morning Shift

Which one of the following species responds to an external magnetic field?

(A) $[Fe(H_2O)_6]^{3+}$ (B) $[Ni(CN)_4]^{2-}$ (C) $[Co(CN)_6]^{3-}$ (D) $[Ni(CO)_4]$

Open solution

Q245. JEE Main 2021 (Online) 20th July Evening Shift

Spin only magnetic moment of an octahedral complex of Fe^{2+} in the presence of a strong field ligand in BM is :

(A) 4.89 (B) 2.82 (C) 0 (D) 3.46

Open solution

Q246. JEE Main 2021 (Online) 20th July Evening Shift

Which one of the following species doesn't have a magnetic moment of 1.73 BM, (spin only value)?

(A) O_2^+ (B) CuI (C) $[Cu(NH_3)_4]Cl_2$ (D) O_2^-

Open solution

Q247. JEE Main 2021 (Online) 20th July Morning Shift

According to the valence bond theory the hybridization of central metal atom is dsp^2 for which one of the following compounds?

(A) $NiCl_2.6H_2O$ (B) $K_2[Ni(CN)_4]$ (C) $[Ni(CO)_4]$ (D) $Na_2[NiCl_4]$

Open solution

Q248. JEE Main 2021 (Online) 20th July Morning Shift

The correct order of intensity of colors of the compounds is :

(A) $[Ni(CN)_4]^{2-} > [NiCl_4]^{2-} > [Ni(H_2O)_6]^{2+}$ (B) $[Ni(H_2O)_6]^{2+} > [NiCl_4]^{2-} > [Ni(CN)_4]^{2-}$

(C) $[NiCl_4]^{2-} > [Ni(H_2O)_6]^{2+} > [Ni(CN)_4]^{2-}$ (D) $[NiCl_4]^{2-} > [Ni(CN)_4]^{2-} > [Ni(H_2O)_6]^{2+}$

Open solution

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Q249. JEE Main 2021 (Online) 18th March Evening Shift

The secondary valency and the number of hydrogen bonded water molecule(s) in $CuSO_4 \cdot 5H_2O$, respectively, are :

(A) 5 and 1 (B) 4 and 1 (C) 6 and 5 (D) 6 and 4

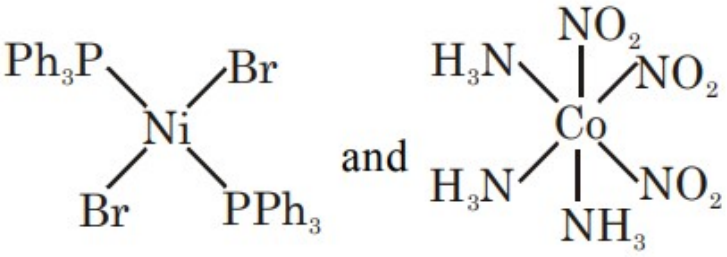
Open solution

Q250. JEE Main 2021 (Online) 18th March Morning Shift

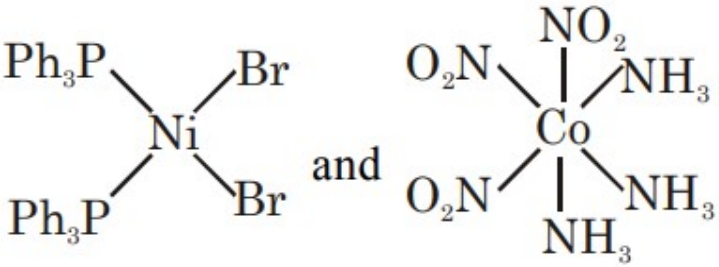
The correct structures of trans-$[NiBr_2(PPh_3)_2]$ and meridional-$[Co(NH_3)_3(NO_2)_3]$, respectively, are

Open solution

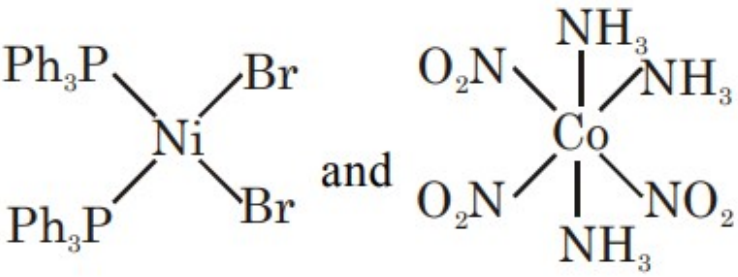
(A)



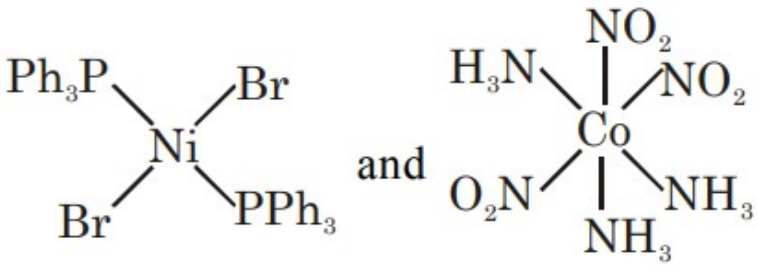
(B)



(C)



(D)



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Q251. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

Match List - I with List - II :

	List - I		List - II
(a)	$[Co(NH_3)_6][Cr(CN)_6]$	(i)	Linkage isomerism
(b)	$[Co(NH_3)_3(NO_2)_3]$	(ii)	Solvate isomerism
(c)	$[Cr(H_2O)_6]Cl_3$	(iii)	Co-ordination isomerism
(d)	$cis-[CrCl_2(ox)_2]^{3-}$	(iv)	Optical isomerism

Choose the correct answer from the options given below :

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(B) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(C) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(D) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

Q252. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

In which of the following order the given complex ions are arranged correctly with respect to their decreasing spin only magnetic moment?

(i) $[FeF_6]^{3-}$

(ii) $[Co(NH_3)_6]^{3+}$

(iii) $[NiCl_4]^{2-}$

(iv) $[Cu(NH_3)_4]^{2+}$

(A) (iii) > (iv) > (ii) > (i)

(B) (ii) > (iii) > (i) > (iv)

(C) (i) > (iii) > (iv) > (ii)

(D) (ii) > (i) > (iii) > (iv)

Q253. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

The hybridization and magnetic nature of $[Mn(CN)_6]^{4-}$ and $[Fe(CN)_6]^{3-}$, respectively are :

(A) sp^3d^2 and diamagnetic

(B) d^2sp^3 and paramagnetic

(C) sp^3d^2 and paramagnetic

(D) d^2sp^3 and diamagnetic

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Q254. JEE Main 2021 (Online) 24th February Evening Shift

The calculated magnetic moments (spin only value) for species $[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MnO_4^{2-} respectively are :

(A) 5.92, 4.90 and 0 BM

(B) 4.90, 0 and 1.73 BM

(C) 5.82, 0 and 0 BM

(D) 4.90, 0 and 2.83 BM

Open solution

Q255. JEE Main 2020 (Online) 6th September Evening Slot

For a d^4 metal ion in an octahedral field, the correct electronic configuration is :

(A) $t_{2g}^4 e_g^0$ when $\Delta_0 < P$

(B) $t_{2g}^3 e_g^1$ when $\Delta_0 > P$

(C) $e_g^2 t_{2g}^2$ when $\Delta_0 < P$

(D) $t_{2g}^3 e_g^1$ when $\Delta_0 < P$

Open solution

Q256. JEE Main 2020 (Online) 6th September Morning Slot

The species that has a spin-only magnetic moment of 5.9 BM, is :
(T_d = tetrahedral)

(A) $[MnBr_4]^{2-}$ (T_d)

(B) $[NiCl_4]^{2-}$ (T_d)

(C) $Ni(CO)_4$ (T_d)

(D) $[Ni(CN)_4]^{2-}$ (square planar)

Open solution

Q257. JEE Main 2020 (Online) 5th September Evening Slot

Consider the complex ions,
trans-$[Co(en)_2Cl_2]^+$ (A) and
cis-$[Co(en)_2Cl_2]^+$ (B).
The correct statement regarding them is :

(A) both (A) and (B) can be optically active.

(B) both (A) and (B) can not be optically active.

(C) (A) can not be optically active, but (B) can be optically active.

(D) (A) can be optically active, but (B) can not be optically active.

Open solution

Q258. JEE Main 2020 (Online) 5th September Morning Slot

The values of the crystal field stabilization energies for a high spin d^6 metal ion in octahedral and tetrahedral fields, respectively, are :

(A) $-0.4\Delta_0$ and $-0.27\Delta_t$

(B) $-1.6\Delta_0$ and $-0.4\Delta_t$

(C) $-0.4\Delta_0$ and $-0.6\Delta_t$

(D) $-2.4\Delta_0$ and $-0.27\Delta_t$

Open solution

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Q259. JEE Main 2020 (Online) 4th September Evening Slot

The one that can exhibit highest paramagnetic behaviour among the following is :

gly = glycinato; bpy = 2, 2'-bipyridine

(A) $[Fe(en)(bpy)(NH_3)_2]^{2+}$

(B) $[Pd(gly)_2]$

(C) $[Co(ox)_2(OH)_2]^-$ ($\Delta_0 > P$)

(D) $[Ti(NH_3)_6]^{3+}$

Open solution

Q260. JEE Main 2020 (Online) 4th September Evening Slot

The Crystal Field Stabilization Energy (CFSE) of $[CoF_3(H_2O)_3]$ ($\Delta_0 < P$) is :

(A) -0.8 Δ_0

(B) -0.4 Δ_0

(C) -0.8 Δ_0 + 2P

(D) -0.4 Δ_0 + P

Open solution

Q261. JEE Main 2020 (Online) 4th September Evening Slot

The molecule in which hybrid MOs involve only one d-orbital of the central atom is :

(A) XeF_4

(B) $[Ni(CN)_4]^{2-}$

(C) $[CrF_6]^{3-}$

(D) BrF_5

Open solution

Q262. JEE Main 2020 (Online) 4th September Morning Slot

The number of isomers possible for $[Pt(en)(NO_2)_2]$ is :

(A) 3

(B) 1

(C) 4

(D) 2

Open solution

Q263. JEE Main 2020 (Online) 4th September Morning Slot

The pair in which both the species have the same magnetic moment (spin only) is :

(A) $[Cr(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$

(B) $[Co(OH)_4]^{2-}$ and $[Fe(NH_3)_6]^{2+}$

(C) $[Mn(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{2+}$

(D) $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$

Open solution

Q264. JEE Main 2020 (Online) 3rd September Evening Slot

The d-electron configuration of $[Ru(en)_3]Cl_2$ and $[Fe(H_2O)_6]Cl_2$, respectively are :

(A) $t_{2g}^4e_g^2$ and $t_{2g}^6e_g^0$

(B) $t_{2g}^6e_g^0$ and $t_{2g}^6e_g^0$

(C) $t_{2g}^6e_g^0$ and $t_{2g}^4e_g^2$

(D) $t_{2g}^4e_g^2$ and $t_{2g}^4e_g^2$

Open solution

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Q265. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

Complex A has a composition of $H_{12}O_6Cl_3Cr$. If the complex on treatment with conc.H_2SO_4 loses 13.5% of its original mass, the correct molecular formula of A is :
[Given: atomic mass of Cr = 52 amu and Cl = 35 amu]

(A) $[Cr(H_2O)_5Cl]Cl_2.H_2O$

(B) $[Cr(H_2O)_4Cl_2]Cl.2H_2O$

(C) $[Cr(H_2O)_3Cl_3].3H_2O$

(D) $[Cr(H_2O)_6]Cl_3$

Q266. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

The complex that can show optical activity is :

(A) cis-$[Fe(NH_3)_2(CN)_4]^-$

(B) trans-$[Cr(Cl_2)(ox)_2]^{3-}$

(C) trans-$[Fe(NH_3)_2(CN)_4]^-$

(D) cis-$[CrCl_2(ox)_2]^{3-}$ (ox = oxalate)

Q267. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

The electronic spectrum of $[Ti(H_2O)_6]^{3+}$ shows a single broad peak with a maximum at 20,300 cm^{-1}. The crystal field stabilization energy (CFSE) of the complex ion, in kJ mol^{-1}, is : (1 kJ mol^{-1} = 83.7 cm^{-1})

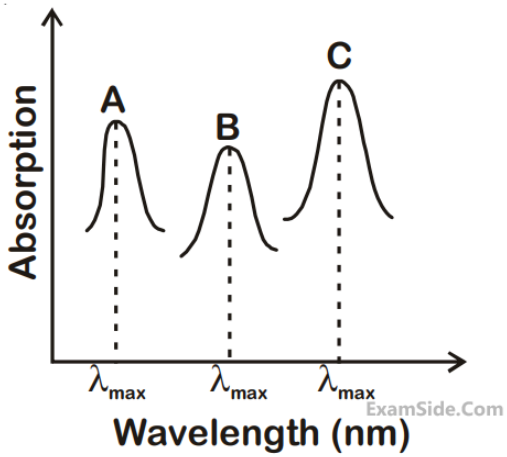
(A) 83.7 (B) 242.5 (C) 145.5 (D) 97

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Q268. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

Simplified absorption spectra of three complexes ((i), (ii) and (iii)) of M^{n+} ion are provided below; their λ_{max} values are marked as A, B and C respectively. The correct match between the complexes and their λ_{max} values is



(i) $[M(NCS)_6]^{(-6+n)}$

(ii) $[MF_6]^{(-6+n)}$

(iii) $[M(NH_3)_6]^{n+}$

(A) A-(i), B-(ii), C-(iii)

(B) A-(ii), B-(iii), C-(i)

(C) A-(ii), B-(i), C-(iii)

(D) A-(iii), B-(i), C-(ii)

Q269. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

The one that is not expected to show isomerism is :

(A) $[Pt(NH_3)_2Cl_2]$

(B) $[Ni(NH_3)_4(H_2O)_2]^{2+}$

(C) $[Ni(en)_3]^{2+}$

(D) $[Ni(NH_3)_2Cl_2]$

Q270. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

For octahedral Mn(II) and tetrahedral Ni(II) complexes, consider the following statements:

(I) both the complexes can be high spin.
(II) Ni(II) complex can very rarely be low spin.
(III) with strong field ligands, Mn(II) complexes can be low spin.
(IV) aqueous solution of Mn(II) ions is yellow in colour.

The **correct** statements are :

(A) (I), (III) and (IV) only

(B) (I) and (II) only

(C) (II), (III) and (IV) only

(D) (I), (II) and (III) only

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Q271. JEE Main 2020 (Online) 2nd September Morning Slot

Consider that a d^6 metal ion (M^{2+}) forms a complex with aqua ligands, and the spin only magnetic moment of the complex is 4.90 BM. The geometry and the crystal field stabilization energy of the complex is

Open solution

(A) tetrahedral and $-1.6\ \Delta_t + 1P$

(B) octahedral and $-2.4\ \Delta_0 + 2P$

(C) tetrahedral and $-0.6\ \Delta_t$

(D) octahedral and $-1.6\ \Delta_0$

Q272. JEE Main 2020 (Online) 9th January Evening Slot

The correct order of the spin-only magnetic moments of the following complexes is :

(I) $[Cr(H_2O)_6]Br_2$

(II) $Na_4[Fe(CN)_6]$

(III) $Na_3[Fe(C_2O_4)_3]$ ($\Delta_0 > P$)

(IV) $(Et_4N)_2[CoCl_4]$

Open solution

(A) (III) > (I) > (II) > (IV)

(B) (II) $\approx$ (I) > (IV) > (III)

(C) (III) > (I) > (IV) > (II)

(D) (I) > (IV) > (III) > (II)

Q273. JEE Main 2020 (Online) 9th January Evening Slot

The isomer(s) of $[Co(NH_3)_4Cl_2]$ that has/have a Cl–Co–Cl angle of 90°, is/are :

Open solution

(A) cis only

(B) cis and trans

(C) meridional and trans

(D) trans only

Q274. JEE Main 2020 (Online) 9th January Morning Slot

$[Pd(F)(Cl)(Br)(I)]^{2-}$ has n number of geometrical isomers. Then, the spin-only magnetic moment and crystal field stabilisation energy [CFSE] of $[Fe(CN)_6]^{n-6}$, respectively, are:

[Note : Ignore the pairing energy]

Open solution

(A) 1.73 BM and $-2.0\ \Delta_0$

(B) 5.92 BM and 0

(C) 2.84 BM and $-1.6\ \Delta_0$

(D) 0 BM and $-2.4\ \Delta_0$

Q275. JEE Main 2020 (Online) 9th January Morning Slot

Complex X of composition $Cr(H_2O)_6Cl_n$ has a spin only magnetic moment of 3.83 BM. It reacts with $AgNO_3$ and shows geometrical isomerism. The IUPAC nomenclature of X is :

Open solution

(A) Hexaaqua chromium (III) chloride

(B) Tetraaquadichlorido chromium(IV) chloride dihydrate

(C) Tetraaquadichlorido chromium (III) chloride dihydrate

(D) Dichloridotetraaqua chromium (IV) chloride dihydrate

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Q276. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

Among (a) – (d) the complexes that can display geometrical isomerism are :

(a) $[Pt(NH_3)_3Cl]^+$

(b) $[Pt(NH_3)Cl_5]^-$

(c) $[Pt(NH_3)_2Cl(NO_2)]$

(d) $[Pt(NH_3)_4ClBr]^{2+}$

(A) (a) and (b) **(B)** (c) and (d) **(C)** (d) and (a) **(D)** (b) and (c)

Q277. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

The correct order of the calculated spin-only magnetic moments of complexes (A) to (D) is:

(A) $Ni(CO)_4$

(B) $[Ni(H_2O)_6]Cl_2$

(C) $Na_2[Ni(CN)_4]$

(D) $PdCl_2(PPh_3)_2$

(A) (C) < (D) < (B) < (A) **(B)** (C) ≈ (D) < (B) < (A)

(C) (A) ≈ (C) ≈ (D) < (B) **(D)** (A) ≈ (C) < (B) ≈ (D)

Q278. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

The complex that can show fac- and mer-isomers is :

(A) $[CoCl_2(en)_2]$ **(B)** $[Pt(NH_3)_2Cl_2]$ **(C)** $[Co(NH_3)_3(NO_2)_3]$ **(D)** $[Co(NH_3)_4Cl_2]^+$

Q279. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

The number of possible optical isomers for the complexes MA_2B_2 with sp^3 and dsp^2 hybridized metal atom, respectively, is :

Note : A and B are unidentate neutral and unidentate monoanionic ligands, respectively.

(A) 0 and 2 **(B)** 0 and 0 **(C)** 0 and 1 **(D)** 2 and 2

Q280. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

Among the statements(a)-(d) the incorrect ones are :

(a) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments.

(b) When Δ_0 < P, the d-electron configuration of Co(III) in an octahedral complex is $t_{2g}^4 e_g^2$

(c) Wavelength of light absorbed by $[Co(en)_3]^{3+}$ is lower than that of $[CoF_6]^{3-}$

(d) If the Δ_0 for an octahedral complex of Co(III) is 18,000 cm^{-1}, the Δ_t for its tetrahedral complex with the same ligand will be 16,000 cm^{-1}

(A) (a) and (b) only **(B)** (b) and (c) only **(C)** (c) and (d) only **(D)** (a) and (d) only

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Q281. JEE Main 2020 (Online) 7th January Morning Slot

The theory that can completely/properly explain the nature of bonding in $[Ni(CO)_4]$ is :

(A) Werner's theory
(B) Valence bond theory
(C) Crystal field theory
(D) Molecular orbital theory

Open solution

Q282. JEE Main 2020 (Online) 7th January Morning Slot

The IUPAC name of complex $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$ is:

(A) Diamminechlorido (methanamine) platinum(II)chloride.
(B) Biasmmine (methanamine)chlorido platinum(II)chloride.
(C) Diamminechlorido (amminomethane) platinum(II) chloride.
(D) Diammine (methanamine) chlorido platinum(II)Chloride.

Open solution

Q283. JEE Main 2019 (Online) 12th April Evening Slot

The coordination numbers of Co and Al in $[Co(Cl)(en)_2]Cl$ and $K_3[Al(C_2O_4)_3]$, respectively, are : (en = ethane-1, 2-diamine)

(A) 3 and 3 (B) 6 and 6 (C) 5 and 3 (D) 5 and 6

Open solution

Q284. JEE Main 2019 (Online) 12th April Evening Slot

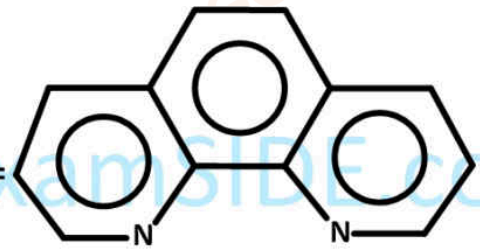
The compound used in the treatment of lead poisoning is :

(A) desferrioxime B (B) Cis-platin (C) D-penicillamine (D) EDTA

Open solution

Q285. JEE Main 2019 (Online) 12th April Morning Slot

The complex ion that will lose its crystal field stabilization energy upon oxidation of its metal to +3 state is :

(Phen =  and

Ignore pairing energy)

(A) $[Fe(phen)_3]^{2+}$ (B) $[Zn(phen)_3]^{2+}$ (C) $[Co(phen)_3]^{2+}$ (D) $[Ni(phen)_3]^{2+}$

Open solution

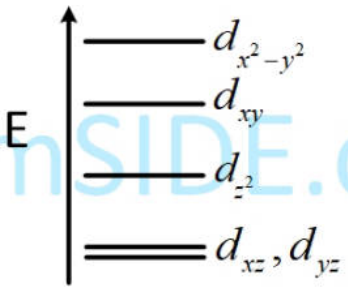
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Q286. JEE Main 2019 (Online) 12th April Morning Slot

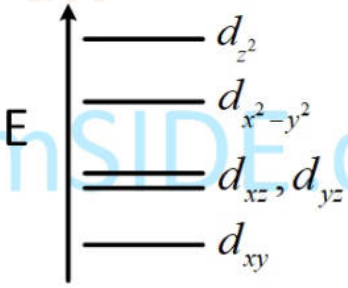
Open solution

Complete removal of both the axial ligands (along the z-axis) from an octahedral complex leads to which of the following splitting patterns? (relative orbital energies not on scale).

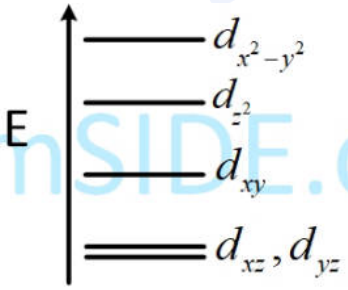
(A)



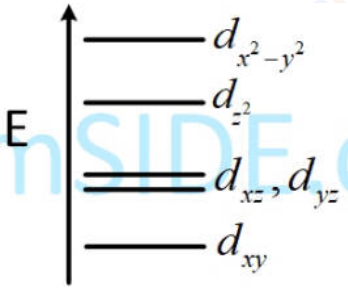
(B)



(C)



(D)



Q287. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

The INCORRECT statement is :

(A) the color of $[CoCl(NH_3)_5]^{2+}$ is violet as it absorbs the yellow light.

(B) the gemstone, ruby, has Cr^{3+} ions occupying the octahedral sites of beryl.

(C) the spin-only magnetic moment of $[Ni(NH_3)_4(H_2O)_2]^{2+}$ is 2.83 BM.

(D) the spin-only magnetic moments of $[Fe(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{2+}$ are nearly similar.

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Q288. JEE Main 2019 (Online) 10th April Evening Slot

The crystal field stabilization energy (CFSE) of $[Fe(H_2O)_6]Cl_2$ and $K_2[NiCl_4]$ respectively, are :

(A) $-0.4\ \Delta_0$ and $-0.8\ \Delta_t$

(B) $-0.6\ \Delta_0$ and $-0.8\ \Delta_t$

(C) $-2.4\ \Delta_0$ and $-1.2\ \Delta_t$

(D) $-0.4\ \Delta_0$ and $-1.2\ \Delta_t$

Open solution

Q289. JEE Main 2019 (Online) 10th April Morning Slot

The species that can have a trans-isomer is :
(en = ethane-1, 2-diamine, ox = oxalate)

(A) $[Cr(en)_2(ox)]^+$ (B) $[Pt(en)Cl_2]$ (C) $[Pt(en)_2Cl_2]^{2+}$ (D) $[Zn(en)Cl_2]$

Open solution

Q290. JEE Main 2019 (Online) 10th April Morning Slot

Three complexes,
$[CoCl(NH_3)_5]^{2+}$(I),
$[Co(NH_3)_5H_2O]^{3+}$ (II) and
$[Co(NH_3)_6]^{3+}$(III)
absorb light in the visible region. The correct order of the wavelength of light absorbed by them is :

(A) (III) > (I) > (II)

(B) (III) > (II) > (I)

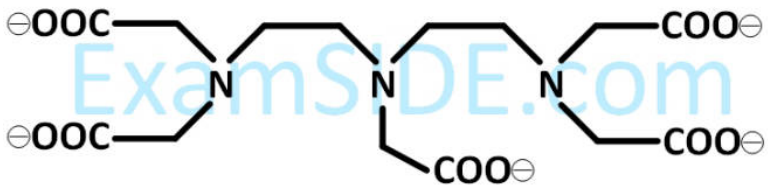
(C) (I) > (II) > (III)

(D) (II) > (I) > (III)

Open solution

Q291. JEE Main 2019 (Online) 9th April Evening Slot

The maximum possible denticities of a ligand given below towards a common transition and inner-transition metal ion, respectively, are :



(A) 8 and 6

(B) 8 and 8

(C) 6 and 6

(D) 6 and 6

Open solution

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Q292. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

The correct statements among I to III are :
(I) Valence bond theory cannot explain the color exhibited by transition metal complexes.
(II) Valence bond theory can predict quantitatively the magnetic properties of transtition metal complexes.
(III) Valence bond theory cannot distinguish ligands as weak and strong field ones.

(A) (II) and (III) only
(B) (I) and (II) only
(C) (I), (II) and (III)
(D) (I) and (III) only

Q293. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

The degenerate orbitals of $[Cr(H_2O)_6]^{3+}$ are :

(A) d_{x^2} and d_{xz}
(B) d_{xz} and d_{yz}
(C) d_{yz} and d_{z^2}
(D) $d_{x^2-y^2}$ and d_{xy}

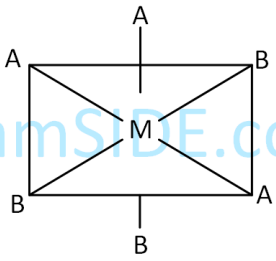
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Q294. JEE Main 2019 (Online) 9th April Morning Slot

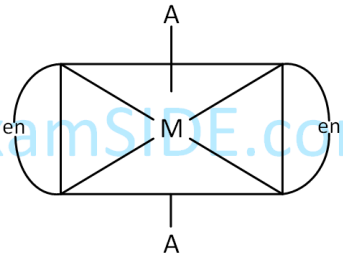
Open solution

The one that will show optical activity is :
(en = ethane-1,2-diamine)

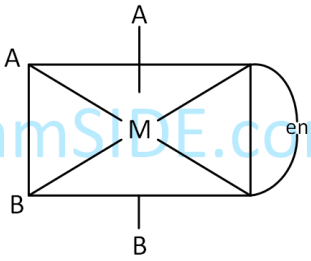
(A)



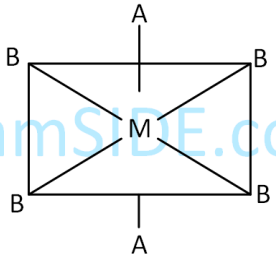
(B)



(C)



(D)



Q295. JEE Main 2019 (Online) 8th April Evening Slot

Open solution

The compound that inhibits the growth of tumors is :

(A) cis-$[Pd(Cl)_2(NH_3)_2]$

(B) trans-$[Pd(Cl)_2(NH_3)_2]$

(C) cis-$[Pt(Cl)_2(NH_3)_2]$

(D) trans-$[Pt(Cl)_2(NH_3)_2]$

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Q296. JEE Main 2019 (Online) 8th April Evening Slot

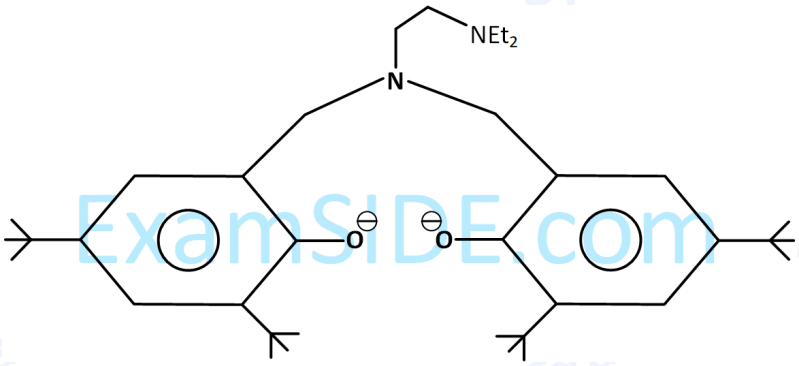
The calculated spin-only magnetic moments (BM) of the anionic and cationic species of $[Fe(H_2O)_6]_2[Fe(CN)_6]$, respectively, are :

(A) 2.84 and 5.92 (B) 4.9 and 0 (C) 0 and 4.9 (D) 0 and 5.92

Open solution

Q297. JEE Main 2019 (Online) 8th April Morning Slot

The following ligand is :



(A) hexadentate (B) bidentate (C) tetradentate (D) tridentate

Open solution

Q298. JEE Main 2019 (Online) 8th April Morning Slot

The correct order of the spin-only magnetic moment of metal ions in the following low-spin complexes, $[V(CN)_6]^{4-}$,$[Fe(CN)_6]^{4-}$, $[Ru(NH_3)_6]^{3+}$, and $[Cr(NH_3)_6]^{2+}$, is :

(A) $V^{2+} > Ru^{3+} > Cr^{2+} > Fe^{2+}$ (B) $V^{2+} > Cr^{2+} > Ru^{3+} > Fe^{2+}$

(C) $Cr^{2+} > V^{2+} > Ru^{3+} > Fe^{2+}$ (D) $Cr^{2+} > Ru^{3+} > Fe^{2+} > V^{2+}$

Open solution

Q299. JEE Main 2019 (Online) 12th January Evening Slot

The magnetic moment of an octahedral homoleptic Mn(II) complex is 5.9 BM. The suitable ligand for this complex is -

(A) CN^- (B) ethylenediamine (C) NCS^- (D) CO

Open solution

Q300. JEE Main 2019 (Online) 12th January Morning Slot

The pair of metal ions that can give a spin only magnetic moment of 3.9 BM for the complex $[M(H_2O)_6]Cl_2$, is -

(A) V^{2+} and Fe^{2+} (B) V^{2+} and Co^{2+} (C) Co^{2+} and Fe^{2+} (D) Cr^{2+} and Mn^{2+}

Open solution

Q301. JEE Main 2019 (Online) 12th January Morning Slot

The metal d-orbitals that are directly facing the ligands in $K_3[Co(CN)_6]$ are -

(A) $d_{x^2-y^2}$ and d_{z^2} (B) d_{xy}, d_{xz} and d_{yz} (C) d_{xz}, d_{yz} and d_{z^2} (D) d_{xy} and $d_{x^2-y^2}$

Open solution

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Q302. JEE Main 2019 (Online) 12th January Morning Slot

$Mn_2(CO)_{10}$ is an organometallic compound due to the presence of -

(A) C–O bond **(B)** Mn – Mn bond **(C)** Mn – O bond **(D)** Mn – C bond

Open solution

Q303. JEE Main 2019 (Online) 11th January Evening Slot

The number of bridging CO ligand(s) and Co-Co bond(s) in $Co_2(CO)_8$, respectively are :

(A) 2 and 0 **(B)** 0 and 2 **(C)** 4 and 0 **(D)** 2 and 1

Open solution

Q304. JEE Main 2019 (Online) 11th January Evening Slot

The coordination number of Th in $K_4[Th(C_2O_4)_4(OH_2)_2]$ is :

($C_2O_4^{2-}$ = Oxalato)

(A) 14 **(B)** 10 **(C)** 8 **(D)** 6

Open solution

Q305. JEE Main 2019 (Online) 11th January Morning Slot

Match the metals (column I) with the coordination compound(s)/enzyme(s) (column II)

(Column I) Metals		(Column II) Coordination compounds(s) enzyme(s)	
(A)	Co	(i)	Wilkinson catalyst
(B)	Zn	(ii)	Chlorophyl
(C)	Rh	(iii)	Vitamin B_{12}
(D)	Mg	(iv)	Carbonic anhydrase

(A) (A)-(iii); (B)-(iv); (C)-(i); (D)-(ii) **(B)** (A)-(iv); (B)-(iii); (C)-(i); (D)-(ii)

(C) (A)-(i); (B)-(ii); (C)-(iii); (D)-(iv) **(D)** (A)-(ii); (B)-(i); (C)-(iv); (D)-(iii)

Open solution

Q306. JEE Main 2019 (Online) 10th January Evening Slot

A reaction of cobalt (III) chloride and ethylenediamine in a 1 : 2 mole ratio generates two isomeric products A (violet coloured) and B (green coloured). A can show optical activity, but B is optically inactive. What type of isomers do A and B represent?

(A) Ionisation isomers **(B)** Linkage isomers

(C) Coordination isomers **(D)** Geometrical isomers

Open solution

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Q307. JEE Main 2019 (Online) 10th January Evening Slot

The difference in the number of unpaired electrons of a metal ion in its high spin and low-spin octahedral complexes is two. The metal ion is :

(A) Mn^{2+} (B) Ni^{2+} (C) Co^{2+} (D) Fe^{2+}

Open solution

Q308. JEE Main 2019 (Online) 10th January Morning Slot

Wilkinson catalyst is : (Et = C_2H_5)

(A) $[(Ph_3P)_3\ RhCl]$ (B) $[(Et_3P)_3RhCl]$ (C) $[(Et_3P)_3IrCl]$ (D) $[(Ph_3P)_3IrCl]$

Open solution

Q309. JEE Main 2019 (Online) 10th January Morning Slot

The total number of isomers for a square planar complex $[M(F)(Cl)(SCN)(NO_2)]$ is

(A) 16 (B) 8 (C) 12 (D) 4

Open solution

Q310. JEE Main 2019 (Online) 9th January Evening Slot

The complex that has highest crystal field splitting energy (Δ), is :

(A) $[Co(NH_3)_5(H_2O)]Cl_3$ (B) $K_2[CoCl_4]$

(C) $[Co(NH_3)_5Cl]Cl_2$ (D) $K_3[Co(CN)_6]$

Open solution

Q311. JEE Main 2019 (Online) 9th January Evening Slot

Homoleptic octahedral complexes of a metal ion 'M^{3+}' with three monodenate ligands L_1, L_2 and L_3 adsorb wavelenths in the region of green, blue and red respectively. The increasing order of the ligands strength is :

(A) $L_3 < L_1 < L_2$ (B) $L_3 < L_2 < L_1$ (C) $L_1 < L_2 < L_3$ (D) $L_2 < L_1 < L_3$

Open solution

Q312. JEE Main 2019 (Online) 9th January Morning Slot

The highest value of the calculated spin only magnetic moment (in BM) among all the transition metal complexes is :

(A) 5.92 (B) 6.93 (C) 3.87 (D) 4.90

Open solution

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Q313. JEE Main 2019 (Online) 9th January Morning Slot

Two complexes $[Cr(H_2O)_6]Cl_3$ (A) and $[Cr(NH_3)_6]Cl_3$ (B) are violet and yellow coloured, respectively. The **incorrect** statement regarding them is :

(A) Δ_0 values of (A) and (B) are calculated from the energies of violet and yellow light, respectively

(B) both are paramagnetic with three unpaired electrons

(C) both absorb energies corresponding to their complementry colors.

(D) Δ_0 value for (A) is less than that of (B).

Open solution

Q314. JEE Main 2018 (Online) 16th April Morning Slot

In Wilkinson's catalyst, the hybridization of central metal ion and its shape are respectively :

(A) sp^3d, trigonal bipyramidal

(B) sp^3, tetrahedral

(C) dsp^2, square planar

(D) d^2sp^3, octahedral

Open solution

Q315. JEE Main 2018 (Online) 16th April Morning Slot

Which of the following complexes will show geometrical isomerism ?

(A) aquachlorobis (ethylenediamine) cobalt (II) chloride

(B) pentaaquachlorochromium (III) chloride

(C) potassium amminetrichloroplatinate (II)

(D) potassium tris(oxalato) chromate (III)

Open solution

Q316. JEE Main 2018 (Offline)

The oxidation states of Cr in $[Cr(H_2O)_6]Cl_3$, $[Cr(C_6H_6)_2]$ and $K_2[Cr(CN)_2(O)_2(O_2)(NH_3)]$ respectively are :

(A) +3, 0 and +4

(B) +3, +4 and +6

(C) +3, +2 and +4

(D) +3, 0 and +6

Open solution

Q317. JEE Main 2018 (Offline)

Consider the following reaction and statements:

$[Co(NH_3)_4Br_2]^+ + Br^- \rightarrow [Co(NH_3)_3Br_3] + NH_3$

(I) Two isomers are produced if the reactant complex ion is a cis-isomer

(II) Two isomers are produced if the reactant complex ion is a trans-isomer

(III) Only one isomer is produced if the reactant complex ion is a trans-isomer

(IV) Only one isomer is produced if the reactant complex ion is a cis – isomer

The correct statements are

(A) (II) and (IV)

(B) (I) and (II)

(C) (I) and (III)

(D) (III) and (IV)

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Q318. JEE Main 2018 (Online) 15th April Evening Slot

The correct order of spin-only magnetic moments among the following is :
(Atomic number : Mn = 25, Co = 27, Ni = 28, Zn = 30)

(A) $[ZnCl_4]^{2-} > [NiCl_4]^{2-} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$

(B) $[CoCl_4]^{2-} > [MnCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

(C) $[NiCl_4]^{2-} > [CoCl_4]^{2-} > [MnCl_4]^{2-} > [ZnCl_4]^{2-}$

(D) $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [NiCl_4]^{2-} > [ZnCl_4]^{2-}$

Open solution

Q319. JEE Main 2018 (Online) 15th April Evening Slot

The total number of possible isomers for square-planar $[Pt(Cl)(NO_2)(NO_3)(SCN)]^{2-}$ is :

(A) 8 (B) 12 (C) 16 (D) 24

Open solution

Q320. JEE Main 2018 (Online) 15th April Morning Slot

The correct combination is :

(A) $[Ni(CN)_4]^{2-}$ – tetrahedral ;

$[Ni(CO)_4]$ – paramagnetic

(B) $[Ni(Cl)_4]^{2-}$ – paramagnetic ;

$[Ni(CO)_4]$ – tetrahedral

(C) $[Ni(Cl)_4]^{2-}$ – square-planar ;

$[Ni(CN)_4]^{2-}$ – paramagnetic

(D) $[Ni(Cl)_4]^{2-}$ – diamagnetic ;

$[Ni(CO)_4]$ –square-planar

Open solution

Q321. JEE Main 2017 (Online) 9th April Morning Slot

$[Co_2(CO)_8]$ displays :

(A) one Co–Co bond, six terminal CO and two bridging CO

(B) one Co–Co bond, four terminal CO and four bridging CO

(C) no Co–Co bond, six terminal CO and two bridging CO

(D) no Co–Co bond, four terminal CO and four bridging CO

Open solution

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Q322. JEE Main 2017 (Offline)

On treatment of 100 mL of 0.1 M solution of $CoCl_3.6H_2O$ with excess $AgNO_3$; 1.2×10^{22} ions are precipitated. The complex is :

(A) $[Co(H_2O)_3Cl_3].3H_2O$

(B) $[Co(H_2O)_6]Cl_3$

(C) $[Co(H_2O)_5Cl]Cl_2.H_2O$

(D) $[Co(H_2O)_4Cl_2]Cl.2H_2O$

Open solution

Q323. JEE Main 2016 (Online) 10th April Morning Slot

Which of the following is an example of homoleptic complex?

(A) $[Co(NH_3)_6]Cl_3$

(B) $[Pt(NH_3)_2Cl_2]$

(C) $[Co(NH_3)_4Cl_2]$

(D) $[Co(NH_3)_5Cl]Cl_2$

Open solution

Q324. JEE Main 2016 (Online) 9th April Morning Slot

Which one of the following complexes will consume more equivalents of aqueous solution of $Ag(NO_3)$?

(A) $Na_3[CrCl_6]$

(B) $[Cr(H_2O)_5Cl]Cl_2$

(C) $[Cr(H_2O)_6]Cl_3$

(D) $Na_2[CrCl_5(H_2O)]$

Open solution

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Q325. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

Identify the correct trend given below : (Atomic No.=Ti : 22, Cr : 24 and Mo : 42)

(A) Δ_o of $[Cr(H_2O)_6]^{2+}$ > $[Mo(H_2O)_6]^{2+}$ and Δ_o of $[Ti(H_2O)_6]^{3+}$ > $[Ti(H_2O)_6]^{2+}$

(B) Δ_o of $[Cr(H_2O)_6]^{2+}$ > $[Mo(H_2O)_6]^{2+}$ and Δ_o of $[Ti(H_2O)_6]^{3+}$ < $[Ti(H_2O)_6]^{2+}$

(C) Δ_o of $[Cr(H_2O)_6]^{2+}$ < $[Mo(H_2O)_6]^{2+}$ and Δ_o of $[Ti(H_2O)_6]^{3+}$ > $[Ti(H_2O)_6]^{2+}$

(D) Δ_o of $[Cr(H_2O)_6]^{2+}$ < $[Mo(H_2O)_6]^{2+}$ and Δ_o of $[Ti(H_2O)_6]^{3+}$ < $[Ti(H_2O)_6]^{2+}$

Q326. JEE Main 2016 (Offline)

Open solution

Which one of the following complexes shows optical isomerism?
(en = ethylenediamine)

(A) cis$[Co(en)_2Cl_2]Cl$

(B) trans$[Co(en)_2Cl_2]Cl$

(C) $[Co(NH_3)_4Cl_2]Cl$

(D) $[Co(NH_3)_3Cl_3]$

Q327. JEE Main 2016 (Offline)

Open solution

The pair having the same magnetic moment is :
[At. No.: Cr = 24, Mn = 25, Fe = 26, Co = 27]

(A) $[Cr(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{2+}$

(B) $[Mn(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{2+}$

(C) $[CoCl_4]^{2-}$ and $[Fe(H_2O)_6]^{2+}$

(D) $[Cr(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$

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Q328. JEE Main 2015 (Online) 11th April Morning Slot

When concentrated HCl is added to an aqueous solution of $CoCl_2$, its colour changes from reddish pink to deep blue.

Which complex ion gives blue colour in this reaction ?

(A) $[CoCl_6]^{4-}$ (B) $[CoCl_6]^{3-}$ (C) $[CoCl_4]^{2-}$ (D) $[Co(H_2O)_6]^{2+}$

Open solution

Q329. JEE Main 2015 (Online) 11th April Morning Slot

Which of the following complex ions has electrons that are symmetrically filled in both t_{2g} and e_g orbitals ?

(A) $[CoF_6]^{3-}$ (B) $[Co(NH_3)_6]^{2+}$ (C) $[Mn(CN)_6]^{4-}$ (D) $[FeF_6]^{3-}$

Open solution

Q330. JEE Main 2015 (Online) 10th April Morning Slot

Which molecule / ion among the following cannot act as a ligand in complex compounds ?

(A) CO (B) CN^- (C) CH_4 (D) Br^-

Open solution

Q331. JEE Main 2015 (Online) 10th April Morning Slot

The correct statement on the isomerism associated with the following complex ions,

(a) $[Ni(H_2O)_5NH_3]^{2+}$,

(b) $[Ni(H_2O)_4\,(NH_3)_2]^{2+}$ and

(c) $[Ni(H_2O)_3\,(NH_3)_3]^{2+}$ is :

(A) (a) and (b) show only geometrical isomerism

(B) (a) and (b) show geometrical and optical isomerism

(C) (b) and (c) show geometrical and optical isomerism

(D) (b) and (c) show only geometrical isomerism

Open solution

Q332. JEE Main 2015 (Offline)

Which of the following compounds is not colored yellow?

(A) $K_3[Co(NO_2)_6]$ (B) $(NH_4)_3[As(Mo_3O_{10})_4]$

(C) $BaCrO_4$ (D) $Zn_2[Fe(CN)_6]$

Open solution

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Q333. JEE Main 2015 (Offline)

The number of geometric isomers that can exist for square planar [Pt (Cl) (py) (NH_3) (NH_2OH)]+ is (py = pyridine) :

(A) 3 (B) 4 (C) 6 (D) 2

Open solution

Q334. JEE Main 2014 (Online) 19th April Morning Slot

An octahedral complex with molecular composition $M.5NH_3.Cl.SO_4$ has two isomers, A and B. The solution of A gives a white precipitate with $AgNO_3$ solution and the solution of B gives white precipitate with $BaCl_2$ solution. The type of isomerism exhibited by the complex is :

(A) Linkage isomerism
(B) Ionisation isomerism
(C) Coordinate isomerism
(D) Geometrical isomerism

Open solution

Q335. JEE Main 2014 (Online) 19th April Morning Slot

Nickel (Z = 28) combines with a uninegative monodentate ligand to form a diamagnetic complex $[NiL_4]^{2-}$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively :

(A) sp^3, two (B) dsp^2, zero (C) dsp^2, one (D) sp^3, zero

Open solution

Q336. JEE Main 2014 (Online) 12th April Morning Slot

Among the following species the one which causes the highest CFSE, Δ_0 as a ligand is :

(A) CN^- (B) NH_3 (C) F^- (D) CO

Open solution

Q337. JEE Main 2014 (Online) 12th April Morning Slot

Which one of the following complexes will most likely absorb visible light ?
(At nos. Sc = 21, Ti = 22, V = 23, Zn = 30)

(A) $[Sc(H_2O)_6]^{3+}$ (B) $[Ti(NH_3)_6]^{4+}$ (C) $[V(NH_3)_6]^{3+}$ (D) $[Zn(NH_3)_6]^{2+}$

Open solution

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Q338. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

Which of the following name formula combinations is **not** correct ?

(A)

	Formula	Name
(1)	$K_2[Pt(CN)_4]$	Potassium tetracyanoplatinate (II)

(B)

	Formula	Name
(2)	$[Mn(CN)_5]^{2-}$	Pentacyanomanganate (II) ion

(C)

	Formula	Name
(3)	$K[Cr(NH_3)_2Cl_4]$	Potassium diammine tetrachlorochromate (III)

(D)

	Formula	Name
(4)	$[Co(NH_3)_4(H_2O)I]SO_4$	Tetraammine aquaiodo cobalt (III) sulphate

Q339. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

Consider the coordination compound, $[Co(NH_3)_6]Cl_3$. In the formation of this complex, the species which acts as the lewis acid is :

(A) $[Co(NH_3)_6]^{3+}$ (B) Cl^- (C) Co^{3+} (D) NH_3

Q340. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

An octahedral complex of Co^{3+} is diamagnetic. The hybridisation involved in the formation of the complex is :

(A) sp^3d^2 (B) dsp^2 (C) d^2sp^3 (D) dsp^3d

Q341. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

The correct statement about the magnetic properties of $[Fe(CN)_6]^{3-}$ and $[FeF_6]^{3-}$ is : (Z = 26).

(A) both are paramagnetic.

(B) both are diamagnetic.

(C) $[Fe(CN)_6]^{3-}$ is diamagnetic, $[FeF_6]^{3-}$ is paramagnetic.

(D) $[Fe(CN)_6]^{3-}$ is paramagnetic, $[FeF_6]^{3-}$ is diamagnetic.

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Q342. JEE Main 2014 (Offline)

The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1, L_2, L_3 and L_4 absorb wavelengths in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is :

Open solution

(A) $L_3 < L_2 < L_4 < L_1$

(B) $L_1 < L_2 < L_4 < L_3$

(C) $L_4 < L_3 < L_2 < L_1$

(D) $L_1 < L_3 < L_2 < L_4$

Q343. JEE Main 2014 (Offline)

The equation which is balanced and represents the correct product(s) is :

Open solution

(A) $[Mg(H_2O)_6]^{2+} + (EDTA)^{4-} \xrightarrow{excess\ NaOH} [Mg(EDTA)]^{2+} + 6H_2O$

(B) $CuSO_4 + KCN \rightarrow K_2[Cu(CN)_4] + K_2SO_4$

(C) $Li_2O + 2KCl \rightarrow 2LiCl + K_2O$

(D) $[CoCl(NH_3)_5]^+ + 5H^+ \rightarrow Co^{2+} + 5NH_4^+ + Cl^-$

Q344. JEE Main 2013 (Online) 25th April Morning Slot

The structure of which of the following chloro species can be explained on the basis of dsp^2 hybridization?

Open solution

(A) $PdCl_4^{2-}$ (B) $FeCl_4^{2-}$ (C) $CoCl_4^{2-}$ (D) $NiCl_4^{2-}$

Q345. JEE Main 2013 (Online) 25th April Morning Slot

In which of the following octahedral complex species the magnitude of Δ_0 will be maximum?

Open solution

(A) $[Co(H_2O)_6]^{2+}$ (B) $[Co(CN)_6]^{3-}$ (C) $[Co(C_2O_4)_3]^{3-}$ (D) $[Co(NH_3)_6]^{3+}$

Q346. JEE Main 2013 (Online) 23th April Morning Slot

The magnetic moment of the complex anion $[Cr(NO)(NH_3)(CN)_4]^{2-}$ is :

Open solution

(A) 5.91 BM (B) 3.87 BM (C) 1.73 BM (D) 2.82 BM

Q347. JEE Main 2013 (Online) 22th April Morning Slot

Which of the following is diamagnetic?

Open solution

(A) $[Fe(CN)_6]^{3-}$ (B) $[Co(ox)_3]^{3-}$ (C) $[FeF_6]^{3-}$ (D) $[CoF_6]^{3-}$

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Q348. JEE Main 2013 (Online) 9th April Morning Slot

Type of isomerism which exists between $[Pd(C_6H_5)_2(SCN)_2]$ and $[Pd(C_6H_5)_2(NCS)_2]$ is :

(A) Linkage isomerism
(B) Coordination isomerism
(C) Ionisation isomerism
(D) Solvate isomerism

Open solution

Q349. JEE Main 2013 (Offline)

Which of the following complex species is not expected to exhibit optical isomerism?

(A) $[Co\ (en)_2\ Cl_2]^-$
(B) $[Co\ (NH_3)_3\ Cl_3]$
(C) $[Co\ (en)\ (NH_3)_2\ Cl_2]^+$
(D) $[Co\ (en)_3]^{3+}$

Open solution

Q350. AIEEE 2012 (Online) 26th May

The complex ion $[Pt(NO_2)(Py)(NH_3)(NH_2OH)]^+$ will give

(A) 2 isomers (Geometrical)
(B) 3 isomers (Geometrical)
(C) 6 isomers (Geometrical)
(D) 4 isomers (Geometrical)

Open solution

Q351. AIEEE 2012 (Online) 19th May

The correct order of ligands in the spectrochemical series is

(A) $Cl^- > en > CN^- > NCS^-$
(B) $CN^- > en > NCS^- > Cl^-$
(C) $NCS^- > CN^- > Cl^- > en$
(D) $en > CN^- > Cl^- > NCS^-$

Open solution

Q352. AIEEE 2012 (Online) 19th May

Square-planar geometry is shown by

(A) $[PtCl_2(NH_3)_2]$
(B) $[NiCl_4]^{2-}$
(C) MnO_4^-
(D) CrO_4^{2-}

Open solution

Q353. AIEEE 2012 (Online) 12th May

Which of the following complex ions will exhibit optical isomerism? (en = 1,2-diamine ethane).

(A) $[Cr(NH_3)_2Cl_2]^+$
(B) $[Co(en)_2Cl_2]^+$
(C) $[Co(NH_3)_4Cl_2]^+$
(D) $[Zn(en)_2]^{2+}$

Open solution

Q354. AIEEE 2012 (Online) 7th May

The d-electron configurations of Cr^{2+}, Mn^{2+}, Fe^{2+} and Co^{2+} are d^4, d^5, d^6 and d^7 respectively. Which one of the following will exhibit the lowest paramagnetic behaviour? (Atomic no. $Cr = 24$, $Mn = 25$, $Fe = 26$, $Co = 27$).

(A) $[Co(H_2O)_6]^{2+}$
(B) $[Cr(H_2O)_6]^{2+}$
(C) $[Mn(H_2O)_6]^{2+}$
(D) $[Fe(H_2O)_6]^{2+}$

Open solution

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Q355. AIEEE 2012

Which among the following will be named as dibromidobis (ethylene diamine) chromium(III) bromide?

(A) $[Cr\ (en)_3\ ]\ Br_3$

(B) $[Cr\ (en)_2\ Br_2\ ]\ Br$

(C) $[Cr\ (en)\ Br_4\ ]^-$

(D) $[Cr\ (en)\ Br_2\ ]\ Br$

Open solution

Q356. AIEEE 2011

Which of the following facts about the complex $[Cr\ (NH_3)_6\ ]Cl_3$ is wrong?

(A) The complex is paramagnetic

(B) The complex is an outer orbital complex

(C) The complex gives white precipitate with silver nitrate solution

(D) The complex involves d^2sp^3 hybridization and is octahedral in shape.

Open solution

Q357. AIEEE 2011

The magnetic moment (spin only) of $[NiCl_4]^{2-}$ is

(A) 5.46 BM

(B) 2.82 BM

(C) 1.41 BM

(D) 1.82 BM

Open solution

Q358. AIEEE 2010

A solution containing 2.675g of $CoCl_3.\ 6NH_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of $AgNO_3$ to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is : (At. Mass of Ag = 108 u)

(A) $[Co(NH_3)_6]Cl_3$

(B) $[CoCl_2(NH_3)_4]Cl$

(C) $[CoCl_3(NH_3)_3]$

(D) $[CoCl(NH_3)_5]Cl_2$

Open solution

Q359. AIEEE 2010

Which one of the following has an optical isomer ?
(en = ethylenediamine)

(A) $[Zn\ (en)\ (NH_3)_2]^{2+}$

(B) $[Co\ (en)_3]^{3+}$

(C) $[Co\ (H_2O)_4\ en]^{3+}$

(D) $[Zn\ (en)_2]^{2+}$

Open solution

Q360. AIEEE 2009

Which of the following has an optical isomer ?

(A) $[Co(en)(NH_3)_2]^{2+}$

(B) $[Co(H_2O)_4(en)]^{3+}$

(C) $[Co(en)_2(NH_3)_2]^{3+}$

(D) $[Co(NH_3)_3Cl]^+$

Open solution

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Q361. AIEEE 2009

Which of the following pairs represents linkage isomers ?

(A) $[Pd\ (P\ Ph_3)_2\ (NCS)_2]$ and $[Pd\ (P\ Ph_3)_2\ (SCN)_2]$

(B) $[Co(NH_3)_5\ NO_3]SO_4$ and $[Co(NH_3)_5\ SO_4]NO_3$

(C) $[Pt\ Cl_2(NH_3)_4]\ Br_2$ and $[Pt\ Br_2(NH_3)_4]\ Cl_2$

(D) $[Cu\ (NH_3)_4]\ [Pt\ Cl_4]$ and $[Pt\ (NH_3)_4]\ [CuCl_4]$

Open solution

Q362. AIEEE 2008

The coordination number and the oxidation state of the element 'E' in the complex $[E(en)_2(C_2O_4)]NO_2$ (where (en) is ethylene diamine) are, respectively,

(A) 6 and 2 (B) 4 and 2 (C) 4 and 3 (D) 6 and 3

Open solution

Q363. AIEEE 2008

In which of the following octahedral complexes of Co (at. no. 27), will the magnitude of Δ_o be the highest?

(A) $[Co(CN)_6]^{3-}$ (B) $[Co(C_2O_4)_3]^{3-}$ (C) $[Co(H_2O)_6]^{3+}$ (D) $[Co(NH_3)_6]^{3+}$

Open solution

Q364. AIEEE 2007

Which one of the following has a square planar geometry?

(A) $[CoCl_4]^{2-}$ (B) $[FeCl_4]^{2-}$ (C) $[NiCl_4]^{2-}$ (D) $[PtCl_4]^{2-}$

Open solution

Q365. AIEEE 2006

Nickel (Z = 28) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[NiX_4]^{2-}$. The number of unpaired electron(s) in the nickel and geometry of this complex ion are, respectively

(A) one, tetrahedral (B) two, tetrahedral

(C) one, square planar (D) two, square planar

Open solution

Q366. AIEEE 2006

The IUPAC name for the complex $[Co(NO_2)(NH_3)_5]Cl_2$ is

(A) nitrito-N-pentaamminecobalt (III) chloride

(B) nitrito-N-pentaamminecobalt (II) chloride

(C) pentaammine nitrito-N-cobalt (II) chloride

(D) pentaammine nitrito-N-cobalt (III) chloride

Open solution

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Q367. AIEEE 2006

In $Fe(CO)_5$, the Fe – C bond possesses :

(A) π-character only
(B) both σ and π characters
(C) ionic character
(D) σ-character

Open solution

Q368. AIEEE 2006

How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion?

(A) Six
(B) Three
(C) One
(D) Two

Open solution

Q369. AIEEE 2005

The oxidation state of Cr in $[Cr(NH_3)_4Cl_2]^+$ is :

(A) +3
(B) +2
(C) +1
(D) 0

Open solution

Q370. AIEEE 2005

The IUPAC name of the coordination compound $K_3[Fe(CN)_6]$ is

(A) Potassium hexacyanoferrate (II)
(B) Potassium hexacyanoferrate (III)
(C) Potassium hexacyanoiron (II)
(D) Tripotassium hexcyanoiron (II)

Open solution

Q371. AIEEE 2005

Which of the following compounds shows optical isomerism?

(A) $[Cu(NH_3)_4]^{+2}$
(B) $[ZnCl_4]^{-2}$
(C) $[Cr(C_2O_4)_3]^{-3}$
(D) $[Co(CN)_6]^{-3}$

Open solution

Q372. AIEEE 2005

The value of the 'spin only' magnetic moment for one of the following configurations is 2.84 BM. The correct one is :

(A) d^4 (in strong ligand field)
(B) d^4 (in weak ligand field)
(C) d^3 (in weak as well as in strong fields)
(D) d^5 (in strong ligand field)

Open solution

Q373. AIEEE 2005

Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour?
(At. No. Cr = 24, Mn = 25, Fe = 26, Co = 27)

(A) $[Cr(CN)_6]^{-3}$
(B) $[Mn(CN)_6]^{-3}$
(C) $[Fe(CN)_6]^{-3}$
(D) $[Co(CN)_6]^{-3}$

Open solution

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Q374. AIEEE 2004

The coordination number of central metal atom in a complex is determined by :

(A) the number of ligands around a metal ion bonded by sigma bonds

(B) the number of only anionic ligands bonded to the metal ion

(C) the number of ligands around a metal ion bonded by sigma and pi- bonds both

(D) the number of ligands around a metal ion bonded by pi-bonds

Open solution

Q375. AIEEE 2004

Which one of the following complexes is an outer orbital complex?

(A) $[Fe(CN)_6]^{4-}$ (B) $[Ni(NH_3)_6]^{2+}$ (C) $[Co(NH_3)_6]^{3+}$ (D) $[Mn(CN)_6]^{4-}$

Open solution

Q376. AIEEE 2004

Coordination compound have great importance in biological systems. In this context which of the following statements is incorrect?

(A) Chlorophylls are green pigments in plants and contains calcium

(B) Carboxypeptidase – A is an enzyme and contains zinc

(C) Cyanocobalamin is B_{12} and contains cobalt

(D) Haemoglobin is the red pigment of blood and contains iron

Open solution

Q377. AIEEE 2004

Which one the following has largest number of isomers? (R = alkyl group, en = ethylenediamine)

(A) $[Ru(NH_3)_4Cl_2]^{+}$ (B) $[Co(en)_2Cl_2]^{+}$

(C) $[Ir(PR_3)_2\ H(CO)]^{2+}$ (D) $[Co(NH_3)_5Cl]^{2+}$

Open solution

Q378. AIEEE 2004

The correct order of magnetic moments (spin only values in B.M.) among is :
(Atomic numbers: Mn = 25; Fe = 26, Co =27)

(A) $[MnCl_4]^{2-} > [CoCl_4]^{-2} > [Fe(CN)_6]^{-4}$ (B) $[Fe(CN)_6]^{-4} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$

(C) $[Fe(CN)_6]^{4-} > [MnCl_4]^{2-} > [CoCl_4]^{2-}$ (D) $[MnCl_4]^{2-} > [Fe(CN)_6]^{4-} > [CoCl_4]^{2-}$

Open solution

Q379. AIEEE 2004

Among the properties (a) reducing (b) oxidising (c) complexing, the set of properties shown by CN^- ion towards metal species is :

(A) c, a (B) b, c (C) a, b (D) a, b, c

Open solution

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Q380. AIEEE 2003

Ammonia forms the complex ion $[Cu(NH_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solutions. What is the reason for it?

(A) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available

(B) In alkaline solutions insoluble $Cu(OH)_2$ is precipitated which is soluble in excess of any alkali

(C) Copper hydroxide is an amphoteric substance

(D) In acidic solutions hydration protects copper ions.

Open solution

Q381. AIEEE 2003

One mole of the complex compound $Co(NH_3)_5Cl_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of $AgNO_3$ solution to yield two moles of AgCl (s). The structure of the complex is :

(A) $[Co(NH_3)_3Cl_3].\ 2NH_3$

(B) $[Co(NH_3)_4Cl_2]\ Cl.\ NH_3$

(C) $[Co(NH_3)_4Cl]\ Cl_2.\ NH_3$

(D) $[Co(NH_3)_5Cl]\ Cl_2$

Open solution

Q382. AIEEE 2003

In the coordination compound, $K_4[Ni(CN)_4]$, the oxidation state of nickel is :

(A) 0 (B) +1 (C) +2 (D) -1

Open solution

Q383. AIEEE 2002

A square planar complex is formed by hybridisation of which atomic orbitals ?

(A) s, p_x, p_y, d_{xy}

(B) s, p_x, p_y, $d_{x^2-y^2}$

(C) s, p_x, p_y, d_{z^2}

(D) s, p_x, p_y, d_{yz}

Open solution

Q384. AIEEE 2002

CH_3 - Mg - Br is an organometallic compound due to :

(A) Mg - Br bond (B) C - Mg bond (C) C - Br bond (D) C - H bond

Open solution

Q385. AIEEE 2002

The most stable ion is :

(A) $[Fe(OH)_3]^{3-}$ (B) $[Fe(Cl)_6]^{3-}$ (C) $[Fe(CN)_6]^{3-}$ (D) $[Fe(H_2O)_6]^{3+}$

Open solution

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Q386. AIEEE 2002

The type isomerism present in nitropentammine chromium (III) chloride is :

(A) optical (B) linkage (C) ionization (D) polymerisation.

Open solution

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Answer Key

Numerical

1. **6**	2. **5**	3. **15**	4. **33**	5. **6**
6. **6**	7. **3**	8. **2**	9. **2**	10. **7**
11. **2**	12. **2**	13. **2**	14. **2**	15. **0**
16. **1**	17. **0**	18. **5**	19. **6**	20. **14**
21. **0**	22. **3**	23. **2**	24. **0**	25. **6**
26. **5**	27. **28**	28. **4**	29. **0**	30. **1**
31. **1**	32. **4**	33. **3**	34. **4**	35. **4**
36. **6**	37. **480**	38. **0**	39. **5**	40. **4**
41. **7**	42. **9**	43. **2**	44. **3**	45. **2**
46. **3**	47. **0**	48. **0**	49. **1**	50. **2**
51. **5**	52. **2**	53. **1**	54. **3**	55. **6**
56. **6**	57. **3**	58. **2**	59. **766**	60. **7**
61. **2**	62. **2**	63. **2**	64. **5**	65. **5**
66. **6**	67. **2**	68. **0**	69. **3**	70. **2**
71. **2**	72. **3**	73. **6**	74. **4**	75. **2**
76. **3**	77. **0**	78. **2**	79. **0**	80. **6**
81. **6**	82. **20**			

MCQ (Single Correct Answer)

83. **D**	84. **C**	85. **A**	86. **A**	87. **C**
88. **B**	89. **A**	90. **A**	91. **D**	92. **A**
93. **B**	94. **D**	95. **C**	96. **D**	97. **C**
98. **C**	99. **D**	100. **C**	101. **D**	102. **C**
103. **D**	104. **B**	105. **A**	106. **C**	107. **B**
108. **D**	109. **C**	110. **B**	111. **A**	112. **C**
113. **A**	114. **D**	115. **B**	116. **A**	117. **C**
118. **B**	119. **B**	120. **A**	121. **C**	122. **C**
123. **A**	124. **A**	125. **D**	126. **C**	127. **C**
128. **A**	129. **C**	130. **D**	131. **C**	132. **C**
133. **A**	134. **D**	135. **A**	136. **C**	137. **B**
138. **A**	139. **A**	140. **B**	141. **A**	142. **D**

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143. **B**	144. **B**	145. **D**	146. **A**	147. **D**
148. **D**	149. **D**	150. **A**	151. **B**	152. **C**
153. **A**	154. **B**	155. **C**	156. **A**	157. **C**
158. **C**	159. **A**	160. **A**	161. **B**	162. **C**
163. **B**	164. **B**	165. **A**	166. **B**	167. **D**
168. **B**	169. **D**	170. **C**	171. **A**	172. **D**
173. **D**	174. **A**	175. **A**	176. **D**	177. **D**
178. **A**	179. **C**	180. **D**	181. **C**	182. **B**
183. **B**	184. **C**	185. **A**	186. **A**	187. **B**
188. **B**	189. **B**	190. **D**	191. **B**	192. **D**
193. **D**	194. **A**	195. **A**	196. **C**	197. **C**
198. **C**	199. **C**	200. **D**	201. **C**	202. **B**
203. **D**	204. **B**	205. **A**	206. **C**	207. **B**
208. **A**	209. **C**	210. **A**	211. **C**	212. **A**
213. **B**	214. **C**	215. **A**	216. **D**	217. **B**
218. **B**	219. **A**	220. **B**	221. **A**	222. **D**
223. **A**	224. **A**	225. **A**	226. **C**	227. **B**
228. **B**	229. **C**	230. **B**	231. **C**	232. **D**
233. **B**	234. **D**	235. **D**	236. **D**	237. **A**
238. **B**	239. **B**	240. **D**	241. **D**	242. **C**
243. **B**	244. **A**	245. **C**	246. **B**	247. **B**
248. **C**	249. **B**	250. **D**	251. **B**	252. **C**
253. **B**	254. **B**	255. **D**	256. **A**	257. **C**
258. **C**	259. **C**	260. **B**	261. **B**	262. **A**
263. **D**	264. **C**	265. **B**	266. **D**	267. **D**
268. **D**	269. **D**	270. **D**	271. **C**	272. **D**
273. **A**	274. **A**	275. **C**	276. **B**	277. **C**
278. **C**	279. **B**	280. **D**	281. **D**	282. **A**
283. **D**	284. **D**	285. **A**	286. **A**	287. **B**
288. **A**	289. **C**	290. **C**	291. **C**	292. **D**
293. **B**	294. **C**	295. **C**	296. **B**	297. **C**
298. **B**	299. **C**	300. **B**	301. **A**	302. **D**
303. **D**	304. **B**	305. **A**	306. **D**	307. **C**

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308. **A**	309. **C**	310. **D**	311. **A**	312. **A**
313. **A**	314. **C**	315. **A**	316. **D**	317. **C**
318. **D**	319. **B**	320. **B**	321. **A**	322. **C**
323. **A**	324. **C**	325. **C**	326. **A**	327. **A**
328. **C**	329. **D**	330. **C**	331. **D**	332. **D**
333. **A**	334. **B**	335. **B**	336. **D**	337. **C**
338. **B**	339. **C**	340. **C**	341. **A**	342. **D**
343. **D**	344. **A**	345. **B**	346. **C**	347. **B**
348. **A**	349. **B**	350. **B**	351. **B**	352. **A**
353. **B**	354. **A**	355. **B**	356. **B**	357. **B**
358. **A**	359. **B**	360. **C**	361. **A**	362. **D**
363. **A**	364. **D**	365. **B**	366. **D**	367. **B**
368. **C**	369. **A**	370. **B**	371. **C**	372. **A**
373. **D**	374. **A**	375. **B**	376. **A**	377. **B**
378. **A**	379. **A**	380. **A**	381. **D**	382. **A**
383. **B**	384. **B**	385. **C**	386. **B**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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